



**Verified Carbon
Standard**

TABREED DISTRICT COOLING GROUP PROJECT



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Client	National Central Cooling Company PJSC - Tabreed
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Summary:

A description of the verification of the project

DNV Business Assurance India Private Limited (hereafter referred to as DNVBAPL) has been commissioned by National Central Cooling Company PJSC - Tabreed (hereafter referred to as “Tabreed” or Project “Proponent”) to verify the greenhouse gas emission reductions reported for project activity “TABREED DISTRICT COOLING GROUP PROJECT” covering the monitoring period from 25-May-2023 to 31-July-2024 (including both the dates) under 07 years renewable crediting period, starting from 25-May-2023 to 24-05-2030 (both dates included), with regard to the relevant requirements for VCS Standard version 4.7.

The proposed project activity is a grouped initiative involving the installation of new district cooling plants with varying capacities to provide cooling services to residential and commercial consumers in Abu Dhabi, United Arab Emirates. The project's objective is to replace the isolated and less efficient cooling technologies with centralized energy efficient units, thereby reducing the electricity consumption and associated CO₂ emissions. The current monitoring period covers the verification for the Project Activity Instance 1: SeaWorld Island (9,000 TR), initially included during the project registration phase and the Project Activity Instance 2: Saadiyat Island (9,000 TR) which has been included for verification within this monitoring period.

The purpose and scope of verification

The purpose of this verification is to have an independent review and ex-post determination of the monitored GHG emission reductions and verify that the monitoring methodology has been implemented according to the registered monitoring plan /3/ and that monitoring data, used to confirm the reductions in anthropogenic emissions by sources is sufficient, definitive and presented in a concise and transparent manner. Specifically, the monitoring plan, monitoring report, and the project's compliance with relevant criteria set forth by VCS, UNFCCC, and the host party have been verified to ensure that the project aligns with the registered design and adheres to the documented conservative assumptions. This verification confirms that the project has been implemented in accordance with the registered project document.

The scope of this verification includes:

- Confirming that the project implementation and operation align with the registered VCS Project Description (PD) /3/.
- Verifying that the implemented monitoring plan is consistent with the registered VCS PD /3/, and the applied baseline and monitoring methodology /6/.
- Ensuring that the actual monitoring systems and procedures conform to those described in the registered monitoring plan.

- Evaluating the GHG emission reduction data and providing a reasonable level of assurance that the reported data is free from material misstatement.
- Verifying that the reported GHG emission data is adequately supported by reliable evidence.

This verification aims to ensure that the reported emission reductions are complete, accurate, and ready for certification.

The monitoring period

Monitoring period: 25-May-2023 to 31-July-2024 (including both the dates)

The method and criteria used for verification

(a) Desk Review, including:

- **Data Completeness Review:** Examination of presented data and information to ensure completeness.
- **Monitoring Plan and Methodology Review:** Detailed assessment of the monitoring plan and methodology, focusing on measurement frequency, metering equipment quality (including calibration requirements), and quality assurance and control procedures.
- **Data Management Evaluation:** Analysis of data management, quality assurance, and control systems to determine their impact on emission reduction generation and reporting.

(b) On-Site Assessment, including:

- **Project Implementation Assessment:** Verification of the proposed VCS project's implementation and operation in alignment with the registered VCS PD.
- **Information Flow Review:** Evaluation of the information flows involved in generating, aggregating, and reporting monitoring parameters.
- **Personnel Interviews:** Interviews with relevant personnel to confirm operational and data collection procedures comply with the monitoring plan in the registered VCS PD.
- **Cross-Check of Information:** Cross-referencing information in the monitoring report with other data sources, such as inventories, purchase records, or similar.
- **Monitoring Equipment Check:** Inspection of monitoring equipment, including calibration performance, and observation of monitoring practices for compliance with requirements of the registered VCS PD, and selected methodology.
- **GHG Data and Emission Reduction Calculations Review:** Examination of calculations and assumptions used in determining GHG data and emission reductions.
- **Quality Control and Assurance Procedures Review:** Identification and review of quality control and assurance procedures established to prevent, detect, and correct any errors or omissions in reported monitoring parameters.

The number of findings raised during the verification

A risk-based approach was employed in conducting this verification.

In the verification process, a total of eleven findings were identified, consisting of **02 Corrective Action Requests (CARs)**, **08 Clarification Requests (CLs)**, and **01 Forward Action Request (FAR)**.

All findings raised have been addressed and resolved by the Project Proponent. The FAR raised during the verification will remain open for the entirety of the crediting period, enabling the assessment team in subsequent reporting cycles to ensure that the FAR requirements relevant to the project activity instance inclusion are effectively addressed and fulfilled.

Any uncertainties associated with the verification

The VCS MR /01/, Emissions Reduction calculations /02/ along with the supporting documents provided are considered to be in line with all the VCS requirements /5/. The verification team has detected no further uncertainties or quality restrictions.

Summary of the verification conclusion

In DNVBAPL's opinion, the emission reductions reported for the project "TABREED DISTRICT COOLING GROUP PROJECT" in the monitoring report are fair and accurately stated. Accordingly, DNVBAPL certifies that the emission reductions achieved by the project during the period 25-May-2023 to 31-July-2024, amount to **22,828 tCO₂** equivalents. The detailed breakdown of emission reductions achieved, categorized by project activity instance and vintage, is presented in Sections 5.2 and 5.3 of this verification report.

DNVBAPL assumes no responsibility for the issuance or utilization of the Verified Carbon Units (VCUs) certified herein. The Project Proponent is responsible for requesting VCU issuance from an approved VCS Program Registry, in accordance with the latest version of the VCS Registration and Issuance Process, specifically the clause on the Project Registration Process. This verification of reported emission reductions is based on the information provided to DNVBAPL. DNVBAPL disclaims any liability or responsibility for any decisions made by persons or entities based on this verification report.

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1 INTRODUCTION

1.1 Objective

DNV Business Assurance India Private Limited (DNVBAPL) has been contracted by National Central Cooling Company PJSC - Tabreed (Project Proponent), to undertake the verification of the project titled “TABREED DISTRICT COOLING GROUP PROJECT” for the monitoring period 25-May-2023 to 31-July-2024 (including both days).

The objective of this verification is to establish that:

- An independent assessment of the project activity was conducted by an accredited verification body, DNV Business Assurance India Private Limited, against the requirements of the current VCS Standard (Version 4.7) /5/ and VCS Program Guide (Version 4.4) /5/ to ensure the quality and consistency of the verification process and report.
- The project activity has been implemented and operated in accordance with the registered project description and the stated guidelines of the applied methodology, AM0117/6/. All physical aspects of the project, including the technology specifications and monitoring protocols are in place as described.
- The newly included project activity instance meets the eligibility requirements specified in the registered VCS Project Description (PD).
- The monitoring report and supporting documentation are transparent, complete, and comprehensive.
- Data has been appropriately recorded and stored as per the applied methodology /6/ and monitoring plan /3/.
- The monitoring system is effectively implemented and fully operational, ensuring the verification of emission reductions without double counting. This also establishes that the reported data is accurate, transparently presented, and free of material errors and omissions.

1.2 Scope and Criteria

The verification of this project is based on the Monitoring Report for the current monitoring period /1/, the registered VCS Project Description (PD) /3/, emission reduction calculation spreadsheets /2/, supporting documentation provided to the verifier /7/-/40/, and information obtained through on-site visit interviews. Additionally, publicly available information was reviewed as necessary and relevant.

Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. Our examination includes an assessment, on a test basis, of evidence relevant to the amounts and disclosures in relation to the project's GHG emission reductions for the defined reporting period.

The verification has been conducted based on the following applicable requirements for this project activity:

- VCS Standard (v4.7) /5a/
- VCS Program Guide (v4.4) /5b/
- CDM Methodology: AM0117, Version 2.0 /6/
- Other relevant rules, including the host country's legislation

The scope of verification, by independent evaluation through examining objective evidence, is as follows.

- Verifying that the project has been implemented as outlined in the VCS PD.
- Assessing compliance with all relevant rules, including those mandated by the host country's legislation.
- Confirming the monitoring system's full functionality to generate accurate voluntary emission reductions without the risk of double counting.
- Ensuring that reported data are accurate, complete, consistent, transparent, and free from material errors or omissions by examining monitoring records and emissions reduction calculations.
- Evaluating the GHG emission reduction data and providing a reasonable assurance opinion on whether it is free from material misstatement.
- Confirming that reported GHG emission data are well-supported by objective evidence.
- Ensuring that the reported emission reductions are comprehensive and accurate to be certified.

The method and criteria used for the verification consisted of the following phases:

1. **Completeness Check and Desk Review:** An initial review to confirm that all necessary documentation is available and complete.
2. **On-site Stakeholder Interviews:** A visit to the project site to engage with stakeholders and verify on-ground practices.
3. **Resolution of Issues and Final Report:** Addressing outstanding issues and issuing a final verification report and VCS Verification Deed of Representation.

All verification activities are conducted under strict impartiality and independence guidelines, with DNVBAPL ensuring that no consulting or recommendations are provided to the client. However, requests for clarifications or corrective actions may offer insights for enhancing monitoring activities.

1.3 Level of Assurance

The verification report is based on the Monitoring report /1/, registered VCS PD /3/, supporting documents /7/-/40/ made available to the verifier, and information collected through performing interviews.

The verification has been planned and organized to achieve a:

- ☒ Reasonable level of assurance as per VCS Standard (v4.7)
- ☐ Limited level of assurance

1.4 Summary Description of the Project

This engagement covers emissions and emission reductions from anthropogenic sources of greenhouse gases included within the project boundary of the following project and period.

Title of Project Activity:	TABREED DISTRICT COOLING GROUP PROJECT
VCS Project ID	4519
Monitoring Period Covered in this Report	25-May-2023 to 31-July-2024 (Including both days)
Project Proponent(s)	NATIONAL CENTRAL COOLING COMPANY PJSC - TABREED
Location of the Project Activity:	The grouped project activity is located in the Abu Dhabi Emirate in the United Arab Emirates. - Project Activity Instance 01 – Sea World DCP, Yas Island Sector YS6 - Project Activity Instance 02 – Saadiyat Island DCP 1, Saadiyat Island SDE 2

The grouped project activity encompasses the installation of high-efficiency district cooling plants designed to serve residential and commercial areas, significantly improving energy efficiency and reducing greenhouse gas (GHG) emissions. The first project activity (PA 01), a 9,000 TR capacity district cooling plant, was installed to provide cooling to residential and commercial consumers in Yas Island. Currently, the project activity is providing a dedicated cooling service to SeaWorld. This project activity was included during the project registration, aiming to replace conventional and less efficient cooling methods. In the current monitoring period, a second project activity (PA 02) has been included, involving the installation of an additional district cooling plant to provide cooling services to both residential and commercial consumers on Saadiyat Island.

Both the activities are implemented and managed by the NATIONAL CENTRAL COOLING COMPANY PJSC - TABREED. By offering cooling energy to residential and commercial consumers, the project displaces the need for traditional, energy-intensive cooling systems, resulting in significant reductions in electricity consumption and subsequent GHG emissions.

Both instances of the project activities (PA 01 and PA 02) have already been successfully commissioned/7/. Details of each plant and its operational status are provided below.

Project Activity Instance	Sea World DCP (PA 01)	Saadiya Island DCP 1 (PA 02)
Commissioning Date	25-May-2023	03-May-2024

Both the project activities have been running satisfactorily since then.

2 VERIFICATION PROCESS

2.1 Method and Criteria

Verification was conducted following DNVBAPL's Process Requirements Procedures, aligned with the specifications in the VCS Requirements, including the VCS Program Guide Version 4.4 and VCS Standard Version 4.7 /5/. The GHG emission reductions were calculated based on the approved baseline and monitoring methodology AM0117: "Introduction of a New District Cooling System," Version 2.0 /6/.

In accordance with ISO 14064-3:2019 and ISO 14065:2020, the assessment team has established a comprehensive process for conducting the verification /8/. This process was meticulously followed throughout the execution of the audit for the project activity, ensuring adherence to the specified standards and guidelines.

The verification process involved below steps.

- **Planning and Notification to VERRA for Site Visit:** The assessment team initiated the planning phase for the GHG programme site visit, commencing with a thorough desk review. A Notice of Validation/Verification Services (NOVS) was digitally submitted via the Verra Project Hub, 15 business days prior to the commencement of the on-site audit.
- **Strategic Analysis:** The assessment team conducted a strategic analysis to gain insights into the project's activities and complexity. This analysis informed the determination of the nature and extent of the verification activities required. The findings from this strategic analysis were utilized to inform the subsequent risk assessment.
- **Risk Assessment:** A comprehensive risk assessment of the GHG statement was performed by the assessment team to identify potential risks of material misstatement or nonconformity with applicable criteria.

- **Evidence Gathering Activities:** Employing a risk-based approach, the assessment team developed a series of evidence-gathering activities designed to collect sufficient and appropriate evidence. This evidence was critical for forming a conclusion and assessing whether the GHG statement adheres to the relevant criteria, considering the principles outlined in the applicable standards and GHG programme.
- **Audit and Sampling Plan:** An audit plan was established that encompassed all necessary sub-elements for an integrated verification process. This plan aligned with the contract requirements, scope, objectives, level of assurance, and materiality considerations. No sampling is applied for the project.
- **Evidence Gathering Plan:** Based on the outcomes of the risk assessment, an evidence-gathering plan was formulated. This plan aimed to mitigate verification risk to an acceptable level and detailed the type and extent of evidence-gathering activities to be conducted.
- **Client Confirmation:** The verification plan was submitted to the client on 19-September-2024 subsequent to the client's confirmation on dates for the site visits on 15th and 16th October-2024.
- **Document Review:** The assessment team conducted a thorough review of relevant documentation, including the verification report, monitoring plan, approved methodology, VCS Project Description (PD), and quality assurance/quality control (QA/QC) procedures. This review ensured that all necessary information was available and aligned with the requirements of the verification process.
- **On-Site Assessment:** The on-site assessment involved direct evaluations of the project scenario, including interviews with key stakeholders and personnel. This phase aimed to validate the implementation of the project activities, gather firsthand insights into operational practices and witness the monitoring of the activity data as per the registered monitoring plan.
- **Resolution of Discrepancies:** Any non-conformities or discrepancies identified during the assessment were promptly addressed and resolved. This process ensured that all findings were documented, and corrective actions were taken where necessary.
- **Independent Review:** An independent technical reviewer conducted a thorough assessment of the findings and conclusions drawn from the verification process.
- **Final Verification Report:** The findings from the document review, on-site assessment, and independent review were compiled into a final verification report. The final verification report and Verification deed of representation are issued.

2.2 Document Review

During the document review phase, DNVBAPL employed standard auditing techniques to assess the quality and integrity of the information provided by the project proponent (PP). The verification process was primarily based on an extensive review of the monitoring report and the associated supporting documentation. This assessment included the following key activities:

- **Completeness Verification:** A thorough examination of the data and information submitted by the PP was conducted to verify their completeness and accuracy.
- **Monitoring Plan and Methodology Review:** The assessment team reviewed the monitoring plan (MP) and monitoring methodology in detail, focusing on critical aspects such as the frequency of measurements, the quality and calibration requirements of metering equipment, and the established QA/QC procedures.
- **Data Management and QA/QC System Evaluation:** An evaluation of the data management processes, and the QA/QC system was undertaken, specifically in relation to their impact on the generation and reporting of emission reductions (ERs).

The initial review of the monitoring report was conducted, after which DNVBAPL requested the PP to provide supporting information and documents referenced as /7/-/40/. These documents were subsequently reviewed to ensure their alignment with the verification requirements. Throughout the verification process, both the revised monitoring report and the supporting documentation were carefully evaluated to confirm the actions taken by the PP in response to the corrective action requests (CARs) and clarification letters (CLs) issued by the verification team.

A comprehensive list of the documents referenced during this verification has been included in Appendix 2.

2.3 Interviews

A site visit to both project activity instances was conducted on 15-October-2024 and 16-October-2024, to verify the information outlined in the table below and to address any issues identified during the document review. The purpose of the site visit was to assess the implementation and operation of the project activity, review relevant evidence, and conduct interviews with key personnel. This process was aimed at confirming the evidence related to project design, implementation, plant operations, environmental impacts, stakeholder engagement, and other critical aspects of the project.

The table below details the on-site interview process, including the personnel interviewed, their respective roles, and the additional information they provided beyond what was included in the project description, monitoring report /1/, and supporting documents. This thorough engagement was instrumental in validating the project's compliance with the established standards and criteria.

1. Sea World DCP - Project Activity Instance 01

Location: Yas Island, Abu Dhabi, United Arab Emirates (UAE)

Date: 15-October-2024

Coverage

- Project design and implementation
- The monitoring procedure,

Source of Information / Persons Interviewed

Mr. Syed Tanveer (Senior Electrical Engineer – Projects), Tabreed

- VCS Monitoring Report and emission reduction calculations,
- Methods and formulae for calculating baseline emissions, project emissions, and leakage
- Technical equipment and operation
- Data uncertainty and residual risks;
- Quality control and quality assurance procedures
- Monitoring equipment including calibration performance
- Implementation of procedures for operations and data collection

Mr. Shashi Prakash (Advisor), Masdar

Mr. Berthly Bitor (Area Superintendent – O&M), Tabreed,

Mr. Pradeep Kumar Gopinathan Nair (Junior Supervisor Operations – O&M), Tabreed

2. Saadiyat Island DCP 1 - Project Activity Instance 02

Location: Saadiyat Island, Abu Dhabi, United Arab Emirates (UAE)

Date: 16-October-2024

Coverage

- Project design and implementation
- The monitoring procedure,
- VCS Monitoring Report and emission reduction calculations,
- Methods and formulae for calculating baseline emissions, project emissions, and leakage
- Technical equipment and operation
- Data uncertainty and residual risks;
- Quality control and quality assurance procedures
- Monitoring equipment including calibration performance
- Implementation of procedures for operations and data collection

Source of Information / Persons Interviewed

Mr. Syed Tanveer (Senior Electrical Engineer – Projects), Tabreed

Mr. Shashi Prakash (Advisor), Masdar

Mr. Berthly Bitor (Area Superintendent – O&M), Tabreed,

Mr. Abhishek Gupta (Senior Supervisor Operations – O&M), Tabreed

2.4 Site Visits

DNVBAPL conducted a comprehensive on-site inspection to verify that all physical features of the project activity, as proposed in the registered Project Description (PD), are in place. The inspection confirmed that the project proponent has operated the project activity and monitored all relevant parameters in accordance with the registered VCS PD during this monitoring period. The verification process maintained a reasonable level of assurance through the following steps:

1. **Implementation and Operation Assessment:** A thorough assessment was conducted to evaluate the implementation and operation of the project activity in accordance with the registered VCS PD.
2. **Information Aggregation and Reporting Review:** The review included an examination of how information regarding monitoring parameters was aggregated and reported.
3. **Personnel Interviews:** Interviews with relevant personnel were conducted to confirm that operational and data collection procedures were implemented in line with the Monitoring Plan (MP). Details are included in section 2.3 above.
4. **Monitoring Equipment Check:** An inspection of the monitoring equipment was carried out, focusing on calibration performance and the observation of monitoring practices against the requirements set forth in the registered VCS PD and the applied monitoring methodology.
5. **Verification of Activity Data:** An on-site data verification activity was conducted to assess the cooling output produced and electricity consumption. This verification involved a detailed comparison of the measured data against the SAP data and ADDC invoices with the objective of ensuring the accuracy and reliability of the reported activity data.
6. **Calculation and Assumption Review:** The assessment included a review of the calculations and assumptions made in determining greenhouse gas (GHG) data and emission reductions (ERs).
7. **QA/QC Procedure Identification:** Finally, the inspection identified the quality assurance/quality control (QA/QC) procedures in place, which are designed to prevent, identify, and correct any errors or omissions in the reported monitoring parameters.

A site visit was conducted for both project activity instances involved in the grouped project activity following the end date of the monitoring period. The details of the sites visited are outlined in the previous section 2.3. Reasonable assurance is achieved through the comprehensive steps described above, which ensure that operational and data collection procedures are in alignment with the Monitoring Plan. These activities collectively contribute to a robust verification process that substantiates the integrity and accuracy of the project's reported outcomes, reinforcing confidence in the validity of the information provided.

2.5 Resolution of Findings

As an outcome of the verification process, the assessment team can raise different types of findings.

In general, where insufficient or inaccurate information is available and clarification or new information is required, the team shall raise a Clarification Request (CL) specifying what additional information is required.

Where a non-conformance arises, the team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- I. The verification is not able to obtain sufficient evidence for the reported emission reductions or part of the reported emission reductions. In this case, these emission reductions shall not be verified and certified.
- II. The verification has identified misstatements in the reported emission reductions. Emission reductions with misstatements shall be discounted based on the verifier's ex-post determination of the achieved emission reductions

The verification process may be halted until this information has been made available to the assessors' satisfaction. Failure to address a CL may result in a CAR. Information or clarifications provided as a result of a CL may also lead to a CAR.

FARs may be raised which are for the benefit of future projects and future verification actors. These have no impact on the completion of the verification activity.

A total of 02 CARs and 08 CLs were raised during the verification process and CL#8 has led into the FAR#11. The same is explained in section 2.5.1 below. Corrective Action Requests and Clarification Requests are detailed in this Verification Report under Annex 3. The Project Proponent is given the opportunity to "close" outstanding CARs and respond to CLs and FARs.

2.5.1 Forward Action Requests

A Forward Action Request (FAR) is issued in instances where certain required deviations or information have not been adequately addressed during the current verification. This FAR ensures that these outstanding issues are carried forward for consideration in subsequent verifications, promoting greater transparency in the verification process. Importantly, the issuance of a FAR does not affect the applicability of the methodology related to the project activity, nor does it impact the additionality, baseline scenario, or the calculation of emission reductions for the project.

The verification team raised **CL#8** regarding the requirement for local stakeholder consultation prior to the inclusion of project activity instance 2, as stipulated in the registered VCS Project Document (PD). The project proponent confirmed that stakeholder consultation was conducted on 19-September-2023 during the validation of project activity instance 1 and not undertaken for PA02 for the following reasons. a) both project activity instances are of similar capacity, b) location is geographically proximate c) PA02 was installed within a close timeframe (8 months) of the stakeholder consultation conducted for PA01, and d) main stakeholders who are the end users and not impacted as they are getting the same service irrespective of the technology and hence the consultation carried out for PA01 was deemed acceptable for PA02. Furthermore, the Project Proponent (PP) sought and obtained deviation approval from VERRA on this matter. VERRA accepted the deviation/41/ and advised the PP to conduct the local stakeholder

consultation prior to the second verification of the project activity. Based on the justification provided by the client, exemption approval from VERRA/41/, and DNVBAPLs technical expertise, the stakeholder consultation conducted for the project activity instance 01 and that the installation and operation of the project activity are expected to enhance the reliability of the cooling services the verification team confirms that no adverse impacts on the consumers of the cooling services are anticipated. As a result, the deviation for project activity instance 2 was accepted.

Also, during the onsite visit, the project proponent was unable to provide a contractual document verifying the requirement of VCU ownership as mentioned in section 3.3 of the MR. However, it was observed that the project proponent has solely invested in and developed the project activity, with the end user having no role or involvement in the project activity itself. The project proponent further clarified that they would include a clause in the customer service agreement specifying the proponent's VCS ownership. Based on this clarification, the VCS ownership of the project proponent is deemed acceptable. However, this clarification led into **Forward Action Request (FAR) #11** and will remain open throughout the crediting period, with the verification team responsible for ensuring that local stakeholder consultations are completed for any future project instances included in the grouped project. Also, the verification team shall check that the clause clarifying the VCU ownership is added in the new customer service agreement for the existing and future project activity instances included in in the grouped project activity. Additionally, the verification team for the second verification shall confirm that the local stakeholder consultation has been completed for PAI 02 prior to the start of the verification process.

2.6 Eligibility for Validation Activities

The grouped project activity falls under Sectoral Scope 1, and DNVBAPL is accredited for the validation and verification of project activities within this scope. However, DNVBAPL did not undertake any validation activities as part of this verification process, nor was DNVBAPL engaged during the validation stage of the project.

3 VALIDATION FINDINGS

3.1 Methodology Deviations

Deviation 1: Grid Emission Factor

Due to the unavailability of data required to calculate the electricity grid emission factor using CDM Tool 7, the project proponent selected a publicly available emission factor from IRENA's 2023 Energy Profile for UAE. This factor (0.457 tCO₂e/MWh) was chosen through comparative analysis for its conservativeness and is applied ex-ante for the first crediting period.

Deviation 2: Benchmarking Boundary

The methodology defines a city boundary as the benchmark location, which the project has interpreted as the administrative region of the Abu Dhabi Emirate due to its distinct jurisdiction within the UAE's seven emirates. This approach ensures compliance with local regulatory boundaries and methodology requirements for the grouped project.

Both deviations were sought, validated, and accepted during the registration stage.

There is no methodology deviation identified during the current monitoring period.

3.2 Project Description Deviations

No deviations from the project description have been observed during the current monitoring period.

3.3 New Project Activity Instances in Grouped Projects

As outlined in the earlier sections, Project Activity Instance 02, Saadiyat Island DCP 1, was included during the current monitoring period based on the eligibility criteria defined in the registered VCS PD. The assessment team conducted a thorough validation of each eligibility condition, substantiated with objective evidence, and additionally verified the applicability requirements of the applied methodology to confirm compliance. It is explained below.

Eligibility/Applicability Criteria	Project Justification	Proponent's	Means of Validation
Applicability of the Methodology (AM0117, version 2.0)			
The methodology is applicable to project activities that reduce CO₂ emissions by means of one, or a combination, of the following measures: (a) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through a new dedicated distribution network; (b) Introduction of new district cooling system(s) that supply cooling to residential and commercial consumers through an existing dedicated distribution network; (c) Expansion of the existing district cooling system(s) by adding a new district cooling plant(s) with or without	Applicable The Project is a new district cooling system that supplies cooling to residential and commercial consumers through a dedicated distribution network.		Project Activity Instance 02, Saadiyat Island DCP 1, involves the construction of a new district cooling plant delivering cooling energy services to residential and commercial consumers via an existing dedicated distribution network. This setup was verified during the onsite visit and further crosschecked through a review of the detailed design report /10/, commissioning reports /15/, and regulatory permit documents /11/-/14/.

Eligibility/Applicability Criteria	Project Justification	Proponent's Means of Validation
expanding a dedicated distribution network. Emission reductions that are gained due to the switch of energy sources shall not be claimed by applying this methodology alone. Therefore, emission reductions due to displacement of the baseline power source can be claimed by the means of the application of this methodology in combination with another relevant approved methodology. In doing so, interactive effects shall be considered as per the "Guidelines for the consideration of interactive effects for the application of multiple CDM methodologies for a programme of activities". For example, if the project district cooling plant is powered partially or completely by a dedicated renewable energy power plant, a project proponent may wish to consider the small-scale methodology 'AMS-I.F: Renewable electricity generation for captive use and mini-grid' to account for emission reductions due to shift of the power source. Otherwise, the most conservative electricity emission factor should be applied to determine emission reductions.	Not applicable The electricity from the grid is the power source in the baseline scenario as well as in the project instance and therefore project will not replace the energy source therefore this condition is not applicable.	The applicability condition does not apply to the project activity since there is no change in energy sources from the baseline scenario. The project activity continues to consume electricity from the grid, as verified during the onsite visit to the project activity instance and through a review of ADDC invoices /16/.

Eligibility condition set in the registered VCS PD

The project activity instances for the grouped project will be located in the geographic boundary of Abu Dhabi Emirates and owned by the project owner or the subsidiary of the project owner.	The project is located in Abu Dhabi. The coordinates are included in the Monitoring report.	The project activity is located within the geographic boundary of the Abu Dhabi Emirate. This has been verified during the on-site visit and confirmed through the Land Site Plan (Affection Plan) issued by the Department
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Eligibility/Applicability Criteria	Project Justification	Proponent's	Means of Validation
All Project Activity Instances shall meet the baseline definition as defined in the respective methodology. The baseline scenario is based on technology penetration.	PA02 meets the eligibility condition.	the	of Municipalities and Transport /11/.
	The latest baseline scenario assessment report is available which has been used during the validation and is applicable for the 3 years of monitoring. The baseline scenario is the same as in registered PD.		The baseline established during project registration remains applicable to project activity instance 02, as validated from the study conducted by Global Market Insights (GMI), page 56/20/. This study determined the penetration of district cooling in Abu Dhabi, finding it to be 18% which is below the threshold of 20% specified in the applied methodology AM0117, version 2.0, Paragraph 19(b).
	PA02 meets the eligibility condition.	the	
	The project activity instance to be included in the grouped project activity is the implementation of a new district cooling plant with or without thermal energy storage.	As per the technical reports and physical installation, it is clear that the project is a new district cooling plant.	
The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology.	PA02 meets the eligibility condition.	the	
	The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology.		
Additionality for this instance has been demonstrated by using requirements	Project instance 2 (PA02) meets the additionality as per the third-party report for the baseline technology assessment. The technology penetration		The project proponent has demonstrated the additionality as per step 1 (clause 5.2.1) of the applied methodology AM0117/6/.

Eligibility/Applicability Criteria	Project Justification	Proponent's	Means of Validation
provided in para 5.2.1; Step 1: Penetration and Performance assessment, in the applied methodology. a. The seasonal energy efficiency ratio (SEER) of the DCP involved in the project activity is higher than a benchmark SEER and (b) The share of the district cooling technologies at the moment of the project registration is less than 20 percent of all cooling technologies within the benchmark boundary in terms of cooling output. The future instances shall comply above requirement of methodology for additionality demonstration. The start date of each project activity instance under the grouped project should not be prior to the start date of the grouped project. The start date of each project activity instance will be determined through documentary evidence. No capacity limits are mentioned in the applied methodology and therefore no capacity limit will be applied for inclusion of new project activity instances. Stakeholder consultation:	is less than 20% and the SEER is higher than the benchmark SEER. PA02 meets the eligibility condition. The PA02 start date is 03-May-2024 as per the commissioning document. PA02 meets the eligibility condition. Not applicable PA02 meets the eligibility condition. The local stakeholder consultation will be	As per paragraph 19a) of the applied methodology, actual operations SEER was observed to be 3.616 which is higher than the benchmark SEER of 2.26. As per paragraph 19b) of the applied methodology, the penetration of district cooling in Abu Dhabi, was found to be 18% which is below the threshold of 20% specified in the applied methodology AM0117, version 2.0. This has been checked and confirmed from the study conducted by Global Market Insights (GMI), page 56 /20/. The start date of the project activity instance 02 is checked and confirmed from the commissioning report of Chiller 3 /15/. The approach adopted is conservative; hence accepted. As per the applied methodology, there is no capacity limit set for the capacity of the project activity. Moreover, the installed capacity of the project activity instance is 9,000 TR which is the same as the capacity of the project activity instance 01 included at the time of project registration. It is observed that local stakeholder consultation wasn't	

Eligibility/Applicability Criteria	Project Justification	Proponent's	Means of Validation
	conducted before the inclusion of any new project instance.		conducted before the inclusion of the project activity instances. PP clarified that both plants are of similar capacity and are geographically proximate, the stakeholders for both project activity instances are considered to be the same. As a result, PP did not conduct a separate stakeholder consultation for project activity instance 2. However, the project proponent has submitted ongoing stakeholder communication records /25/ which the assessment team checked and confirmed. Additionally, PP) has sought and obtained exemption approval/41/ for this deviation from VERRA. Therefore, this deviation for project activity instance 2 is accepted. However, FAR#11 is raised, and it will remain open for the rest of the crediting period. The verification team, in future monitoring periods, shall verify that local stakeholder consultation is completed before the inclusion of all future project activity instances to the grouped project.
Environmental Impact:	Every new project instance will follow the host country's environmental regulations related to		The project has provided the environmental approval from EAD /26/ in addition to approvals from ADPC Ports, DCT, ADNOC, ADAC Air Navigation

Eligibility/Applicability Criteria	Project Justification	Proponent's	Means of Validation
environmental approvals.			Services, ADDC, ADM Assets, ADSSC, and TRANSCO.
Sampling:	No sampling is used for the emission reduction calculation. All the project instances shall have the monitoring parameters as explained in the monitoring section of the methodology AM0117 version 02	No sampling is applied to the project activity. Both the sites are visited by the assessment team and complete monitored data of cooling output and electricity consumption from SAP records /17/, /18/, and ADDC invoices /16/.	
Conditions to avoid double counting - Unique Identification:	All the new project instances will be named uniquely based on location as PI 1 is called 'Sea World District Cooling Plant'. To prevent double counting of District Cooling Plants under the project, District cooling plants shall be uniquely marked, and new project instance implementers shall provide assurances/ declarations that they are not included in any other CDM or Voluntary GHG offset schemes outside of this project.	Saadiyat Island DCP 1 is uniquely named PA02 while included in the grouped project activity. Also, the project proponent has provided the declaration /24/ confirming that. - The project was neither registered nor rejected in any other GHG programme other than VERRA. - The project is a voluntary activity that is not covered under any national/local regulations in UAE. - The project will not claim any GHG benefits in other GHG schemes other than VERRA.	This is found appropriate and hence accepted.
<u>Programme management- Approval of new project instance by project proponent; collection of data and record keeping:</u>	The project proponent approves each project instance to be included in its registered group		During the site visit to the project activity instance 02, it is observed the project proponent Tabreed has maintained all

Eligibility/Applicability Criteria	Project Justification	Proponent's	Means of Validation
	project activity. Project proponent Tabreed will be responsible for keeping all the emission reduction data, including the details of project instances start date, commissioning document, and other required information.		project-related information/documents such as regulatory approvals documents /10/-/14/, commissioning documents /15/, monitoring data records /16/-/19/, metering details /21/, calibration records/22/, /23/ etc.
Conditions to provide an affirmation that funding from Annex I parties, if any, does not result in a diversion of official development assistance.	Under this project, new project instance operators shall be required to provide an affirmation that funding from annex I parties, if any, does not result in a diversion of official development assistance.		The project proponent has provided the declaration /24/ confirming that - The project has not obtained any ODA funding for the project activity This is found appropriate and hence accepted.
VCU ownership	End users receiving Cooling demand under the specific new project instance contractually cede their rights to claim and own emission reductions under the VCS to the project proponent of the group project.		During the onsite visit, the project proponent was unable to provide a contractual document verifying this requirement. However, it was observed that the project proponent has solely invested in and developed the project activity, with the end user having no role or involvement in the project activity itself. The project proponent further clarified that they would include a clause in the customer service agreement specifying the proponent's VCS ownership. Based on this clarification, the VCS ownership of the project proponent is deemed

Eligibility/Applicability Criteria	Project Justification	Proponent's Means of Validation
Awareness and agreement of those operating a new project instance on group project subscription	Contractual provisions to ensure that those operating the new project instance are aware and have agreed that their activity is being subscribed to the new project instance.	acceptable. However, CAR#8 is extended to ensure that future verifications confirm the inclusion of this clause in customer agreements for all current and future project activity instances under the grouped project activity. The Project Activity Instance 02 is owned and operated by the project proponent Tabreed. The project ownership is checked from authority approval and commissioning documents /15/.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 VERIFICATION FINDINGS

4.1 Project Details

Item	Evidence-gathering activities, evidence checked, and assessment conclusion:				
Audit history	Audit Type	Monitoring Period	Program	VVB	Number of Years
	Verification (Current)	25-May-2023 to 31-July-2024	VCS	DNVBAPL	1.18

	In accordance with ISO 14064-3:2019 /27/, section 6.1.3 and the VCS requirements /5/, the assessment team conducted a thorough desk review during the pre-engagement phase to validate the duration of the current monitoring period, and the crediting period for this project activity. Since this is the first monitoring period, there is no verification history for the project. Additionally, the VVB verified the validation audit history by conducting personnel interviews and focused group discussions with the project proponent during the onsite audit. Evidence Checked: Registered PCS PD version 01.9 dated 10-May-2024 /3/, Validation Report version 1.2 dated 20-May-2024 /4/, and Project Webpage https://registry.terra.org/app/projectDetail/VCS/4519 DNVBAPL concludes that the activities performed, and the evidence reviewed confirm the audit history in alignment with ISO 14064-3 /27/ and VCS Standard Version 4.7 /5/ requirements. These findings are deemed accurate and satisfactory by the assessment team.
Double counting and participation in other GHG programs	The project is registered solely under VCS with project ID 4519. It has not been submitted under any other GHG program for the current monitoring period, from 25-May-2023 to 31-July-2024 (inclusive). This was verified through checks on the VERRA registry website, along with similar searches for other registries using the project title, capacity, and project proponents' details. No matching project was identified in any of these registries or similar mechanisms. Additionally, the assessment team conducted an independent search to identify any potential registrations or claims of GHG-related benefits for the current monitoring period. Based on these checks and the declaration submitted by the project proponent /24/, the assessment team confirms that there is no double counting for this project activity in the current monitoring period.
No double claiming with emissions trading programs or binding emission limits	The project is not part of any emissions trading program or any mechanism involving GHG allowance trading. The project proponent (PP) has submitted a declaration /24/ stating that <i>"The project is voluntary activity doesn't cover under any national/local regulations in UAE"</i>. Additionally, the assessment team confirms that there is no emission trading program existing in UAE and no binding emission limit applies.

No double claiming with other forms of environmental credit	The project does not seek or receive any other form of GHG-related environmental credits. The assessment team cross-verified issuance records on the Verra website, confirming that emission reductions generated from the project activity are not and will not be double-counted. Additionally, the project proponent provided the declaration /24/ which states that <i>“The project will not claim any GHG benefits in other GHG schemes other than VERRA.”</i>
Supply chain (scope 3) emissions double claiming.	The assessment team confirms that the project proponent, a district cooling service provider, supplies cooling or chilled water produced through district cooling plants, and that the associated greenhouse gas (GHG) emissions are affected by the project activities. The project proponent has publicly disclosed that <i>“Carbon credits may be issued through Verified Carbon Standard project ID 4519 - TABREED DISTRICT COOLING GROUP PROJECT for the greenhouse gas emission reductions or removals associated with National Central Cooling Company PJSC – Tabreed’s cooling/chilled water produced through the district cooling plants.”</i> in accordance with paragraph 3.24.7 of the VCS Standard v4.7. The assessment team has verified this disclosure by reviewing the company’s official sustainability webpage: https://www.tabreed.ae/sustainability/ (last accessed on 17 April 2025).
Sustainable development contributions	In accordance with VCS Standard Version 4.7 /5/, section 3.17.1, the project proponent is required to “demonstrate that a project contributes to at least three SDGs by the end of the first monitoring period and in each subsequent monitoring period.” The current verification covers the second monitoring period, from 25-May-2023 to 31-July-2024 (inclusive). For this period, the project proponent is voluntarily demonstrating contributions to three SDGs, as detailed below: SDG target 13; Total greenhouse gas emissions avoided per year The project activity involved implementing energy-efficient cooling technologies in order to provide cooling services to residential and commercial consumers, resulting in an emission reduction of 22,828 tCO₂e for the current monitoring period from 25-May-2023 to 31-July-2024 (inclusive).

	The assessment team reviewed the emission reduction calculations for this period in the ER sheet /02/ and verified the application of correct formulas as per the applied methodology /6/. The team also confirmed the accuracy and completeness of the monitored data using SAP records /17/-/19/ and ADDC invoices /16/. Therefore, the assessment team concludes that the emission reduction calculations are accurate, authentic, and consistent with the registered PD /3/ and the applied methodology /6/. In addition to SDG 13, the project activity is also contributing to SDG 8 and SDG 11 which is explained as below. Section 1.12 of the revised MR/1 specifies that a minimum of three operators are employed during both the construction and operational phases of the project activity. The assessment team has verified compliance with this requirement during the site visit and through interviews conducted with the plant operations team. Additionally, the project proponent has claimed that the implementation of the project activity has contributed to a reduction in air pollutants. The assessment team concurs with this assertion, noting that the project activity has effectively reduced energy consumption. This reduction minimizes reliance on grid-generated power, which would have resulted in higher emissions of air pollutants. To substantiate the appropriateness of this claim, the assessment team reviewed and validated the article “Benefits of District Cooling in the Context of the Middle East Region” /29/, which aligns with the observed environmental benefits of the project activity.
Additional information relevant to the project	The assessment team confirms that, according to the information provided in the monitoring report, no commercially sensitive information has been excluded from the public versions of the project documents. This is in full compliance with the VCS Program requirements regarding permissible exclusions. The information was further verified through interviews with the project proponent’s representative.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Stakeholder identification	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/ and onsite interviews with relevant personnel /9/ Assessment Conclusion: The stakeholders identified and documented during the validation remain unchanged. The project proponent didn't conduct a separate stakeholder consultation for project activity instance 02 as per eligibility criteria 8 for the inclusion of project activity instance in the grouped project. Considering both the plants are of similar capacity and geographically proximate, the assessment team has accepted this deviation as this does not impact the applicability of the methodology related to the project activity, nor does it impact the additionality, baseline scenario, or the calculation of emission reductions for the project. Furthermore, The PP sought exemption approval from VERRA on this matter and VERRA accepted this deviation and advised PP to conduct the stakeholder consultation before second verification of the project activity. Furthermore, PP sought and obtained deviation approval from VERRA on this matter. VERRA accepted the deviation/41/ and advised the PP to conduct the local stakeholder consultation prior to the second verification of the project activity. Accordingly, FAR#11 has been raised for the upcoming verification period to ensure that the verification team can confirm the completion of stakeholder consultation for Project Activity Instance (PAI) 02 before the commencement of the second verification. Additionally, the verification team can confirm that local stakeholder consultation is conducted before any new project activity instances are included in the grouped project activity. As this is the first verification for the project activity, the assessment team can confirm that no new stakeholders have been identified during the current period. The stakeholder consultation was conducted during the validation stage on 19-September-2023 via MS Teams. The revised Monitoring Report (MR) complies with the MR completion guidelines, (template) version 4.4 and Section 3.18 of the VCS Standard v4.7.
Legal or customary tenure/access rights	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/ and onsite interviews with relevant personnel /9/ Assessment Conclusion: The grouped project activity is owned and operated by the project proponent which checked and confirmed from

	the project design /10/, regulatory approvals /11/-/14/, and commissioning reports /15/.
Stakeholder diversity and changes over time	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/ and onsite interviews with relevant personnel /9/ Assessment Conclusion: Since there is no change in stakeholders from the validation stage of the project, it is not applicable.
Expected changes in well-being	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/ and onsite interviews with relevant personnel /9/ Assessment Conclusion: The efficiency improvements gained from district cooling systems provide multiple environmental and operational benefits. By optimizing cooling efficiency, these systems help reduce emissions of pollutants and greenhouse gases, significantly lowering the carbon footprint of urban areas. They also use fewer refrigerant gases, which have high global warming and ozone-depleting potential, further benefiting the environment. /29/ Additionally, district cooling systems reduce environmental noise and vibrations, enhancing the quality of life for nearby residents. With centralized cooling infrastructure, they minimize the need for individual air conditioning units, leading to quieter neighborhoods and less energy demand across the grid.
Location of stakeholders	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/ and onsite interviews with relevant personnel /9/. Assessment Conclusion: Since the majority of the stakeholders involved in the project activity—comprising employees, government and approving authorities, as well as local communities and consumers—are located within the geographic boundaries of Abu Dhabi Emirate, this was observed during the onsite interviews.
Location of resources	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/ and onsite interviews with relevant personnel /9/

Assessment Conclusion: The primary resource used in the project activity is electricity supplied by the Abu Dhabi state-owned utility, as verified and confirmed through ADDC invoices /16/.

4.2.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Ongoing consultation	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD/3/, onsite interviews with relevant personnel /9/, Tabreed Enquiry Process and Customer Service Activity Report/25/. Assessment Conclusion: As checked and verified from Tabreed's Customer Service Charter outlines the complaint handling process, detailing how complaints are lodged, tracked, and resolved. Lodging Complaints: - End-users can lodge complaints by calling 800 TASLEEM, emailing customerservice@tasleem.ae, or visiting the Tasleem website. - Bulk-offtake customers can call 800 TABREED, email customerservice@tabreed.ae, or contact their Account Manager. Acknowledgment: Complaints are acknowledged via phone or email on the same business day or the next if submitted during weekends or holidays. Accessibility: Customers can access copies of their complaint logs by contacting customer service during business hours. Additionally, the assessment team reviewed the customer service activity report /25/, which logs all inquiries and complaints received, along with Tabreed's inquiry process /25/, which outlines the complaint handling workflow. Based on this review, the team concluded that the ongoing communication process is appropriately designed to meet the ongoing communication requirements and effectively supports customer interaction and issue resolution.
Date(s) of stakeholder consultation	Local stakeholder consultation was carried out on 19-September-2023 as checked and verified from local stakeholder consultation documents /28/.

Communication of monitored results	Monitored results can be requested by emailing customerservice@tasleem.ae or calling 800 TASLEEM (8275336) during regular business hours, as observed during the site visit.
Consultation records	Consultation records are documented in the customer service activity report /25/ and a sample report is provided to the assessment team.
Stakeholder input	The stakeholder input process is continuous, with all comments/inquiries/complaints recorded and addressed appropriately within the customer service activity report /25/. This ensures that stakeholder feedback is systematically documented and responded to in a timely and organized manner.

4.2.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	The project has obtained all the necessary approvals from the relevant authorities, with the approval documents /11/-/14/ thoroughly reviewed and validated by the assessment team.
The outcome of the FPIC discussion	The land for the project activity was allocated by the Department of Municipalities and Transport, as verified through the Land Site Plan (Affection Plan) /11/. Additionally, the site location was cross-checked using Google Earth imagery to ensure no communities were residing on the land prior to the project, confirming that no land encroachment or displacement occurred. Based on this evidence, the assessment team concludes that the project activity has not infringed on any occupied land or necessitated relocation of people.

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, onsite interviews with relevant personnel /9/, Tabreed Enquiry Process and Customer Service Activity Report /25/.

	Assessment Conclusion: No grievance was received for the project activity during the current monitoring period as checked and confirmed from Customer Service Activity Reports /25/.
Grievance redress procedure	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, onsite interviews with relevant personnel /9/, Tabreed Enquiry Process and Customer Service Activity Report/25/ Assessment Conclusion: As checked and verified from Tabreed's Customer Service Charter outlines the complaint handling process, detailing how complaints or grievances are lodged, tracked, and resolved. Additionally, the assessment team reviewed the customer service activity report /25/ for the monitoring period, which logs all inquiries, complaints, and grievances received in line with Tabreed's inquiry process /25/, which outlines the complaint/grievance handling workflow. Based on this review, the team concluded that the Tabreed inquiry process is appropriately designed to meet the grievance redress procedure requirements and effectively supports customer interaction and issue resolution. Additionally, Tabreed has developed an employee grievance policy that follows three stages of escalation for robust governance as checked and verified from page 14 in Tabreed's 2023 ESG report /32/ contributed to SDG 8.8.

4.2.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
The project was open for a 30-day listing period during the validation stage and no public comments were received during that period.	Not applicable	Not applicable

4.2.6 Risks to Local Stakeholders and the Environment

4.2.6.1 Management Experience

Evidence gathering activities were performed through the desk review and onsite assessment. The assessment team has checked the evidence such as Registered VCS PD /3/, Tabreed company information/31/, and performed onsite interviews with relevant personnel /9/.

Assessment Conclusion: The assessment team confirms that Tabreed possesses extensive management experience to successfully support and operate the project. Established in 1998 as a publicly listed company, Tabreed has been providing cooling services to residential and commercial customers for over two decades /31/. With 86 plants across the GCC, Tabreed delivers cooling solutions to landmark projects including The Louvre Abu Dhabi, Downtown Dubai, Sheikh Zayed Grand Mosque, Ferrari World Abu Dhabi, Yas Water World, and Dubai Metro, among others. Based on this proven expertise, the assessment team has confidence in Tabreed's recruitment and operational strategy to ensure the necessary experience and capabilities for successful project execution.

4.2.6.2 Risk Assessment

Evidence gathering activities were performed through the desk review and onsite assessment. The assessment team has checked the evidence such as Registered VCS PD /3/, Benefits of district cooling in the context Middle East region /29/, Tabreed ESG Report 2023 /32/ prepared in accordance with the GRI 2021 standards, and performed onsite interviews with relevant personnel /9/.

The assessment conclusion for each of the risk categories is included in the below table.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to Stakeholders' wellbeing	No risk has been identified. The project activity involves installing an energy-efficient district cooling plant to deliver cooling services to residential and commercial consumers. This initiative brings significant environmental and community benefits /29/, as outlined in Section 4.2.1 of this verification report.
Risks to stakeholder participation	No risk has been identified. The project activity involves the operation of a district cooling plant, for which the project proponent brings over 25 years of proven experience /31/. The plant is managed by skilled operators, and with the proponent's extensive expertise in district cooling operations as described in section 4.2.6.1 above, the assessment team concludes that there is no risk to effective stakeholder participation.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Working conditions	No risk has been identified. As a publicly listed company /31/, /33/, Tabreed adheres to UAE labor laws, ensuring fair treatment and respectful working conditions for its employees. This compliance has been confirmed through a review of page 40 in Tabreed's 2023 ESG report /32/ which adheres to GRI 2021 requirements.
Safety of women and girls	No risk has been identified. Tabreed protects labour rights and promotes safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment as validated on page 14 in Tabreed's 2023 ESG report /32/ contributed to SDG 8.8.
Safety of minority and marginalized groups, including children	No risk has been identified. The project activity entails installing an energy-efficient district cooling plant to provide cooling services to residential and commercial consumers, with no adverse impact on minority or marginalized groups, including children.
Pollutants (air, noise, discharges to water, generation, and release of hazardous materials and chemical pesticides and fertilizers	No risk has been identified. Tabreed complies and strictly adheres to trade effluent discharge regulations mandated by the relevant statutory authorities in each emirate as checked and verified from page 30 in Tabreed's 2023 ESG report /32/. These regulations include: - The Trade Effluent Control Regulations 2022 was issued by the Department of Energy for Abu Dhabi, Al Ain, and other emirates. - Local Order No. 61 of 1999 concerning Environmental Standards and Allowable Limits of Pollutants on Land, Water, and Air Environment issued by Dubai Municipality for Dubai. - The Trade Effluent Disposal Waiver Letter was issued by Dubai Municipality on 27-June-2011.

4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Evidence gathering activities were performed through the desk review and onsite assessment. The assessment team has checked the evidence such as Registered VCS PD /3/, Tabreed ESG Report 2023 /32/ prepared in accordance with the GRI 2021 standards, Federal Decree-Law No. (33) of 2021 /34/, and performed onsite interviews with relevant personnel /9/.

The assessment conclusion for each of the risk categories is included in the below table.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	No risk has been identified. Tabreed is committed to fostering a diverse and inclusive workforce, viewing it as essential to its business strategy, as highlighted on page 34 of its ESG report /32/. Additionally, the company strives to eliminate all forms of discrimination, a goal confirmed by information on pages 13 and 17 of the same report /32/. As a publicly listed company, Tabreed adheres to UAE labour laws, including Federal Decree-Law No. (33) of 2021, Article (4) which mandates equality and non-discrimination, as outlined on page 5 of the decree.
Sexual harassment	No risk has been identified. Tabreed complies with UAE labour laws, including Federal Decree-Law No. (33) of 2021. Article (14) of this decree mandates the prohibition of forced labour and addresses other protections, including the prohibition of sexual harassment, bullying, and any form of verbal, physical, or psychological violence against workers by employers, superiors, colleagues, or other individuals in the workplace, as detailed on page 10 of the decree.
Gender equity in labor and work	No risk has been identified. As highlighted above, Tabreed prioritizes diversity, equity, and inclusion, as evidenced by its ESG report (page 34) /32/. The report shows notable progress in gender diversity, with female representation increasing from 7% to 10% between 2021 and 2023. Additionally, there has been a significant reduction in the gender-based pay gap, reflecting Tabreed's commitment to equitable workplace practices.

Forced labor	No risk has been identified. As outlined above, Tabreed adheres to UAE labour laws, including Federal Decree-Law No. (33) of 2021. Article (14) of this decree strictly prohibits forced labour, stating that an employer must not use any means to compel a worker to work against their will, nor threaten penalties to enforce compliance. This law ensures that no employee is obligated to perform work or provide services involuntarily, as detailed on page 10 of the decree.
Child labor	No risk has been identified. As per page 17 of the ESG report 2023 /32/, Tabreed ensures compliance with International Labour Organization (ILO) Conventions, specifically Conventions 138 & 182 on child labour and Conventions 29 & 105 on forced labour. The company also adheres to Principle 5 of the United Nations Global Compact (UNGC), promoting the effective abolition of child labour. Additionally, Tabreed mandates that its suppliers implement effective child labour policies, as confirmed on page 23 of the ESG report /32/.
Human trafficking	No risk has been identified. UAE laws strictly prohibit all forms of human trafficking. As outlined above, the project proponent's policy on human rights aligns with ILO conventions, as verified on page 40 of the ESG report /32/. Furthermore, Tabreed requires its suppliers to implement effective human rights policies, as confirmed on page 23 of the ESG report /32/.

4.2.7.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk has been identified.	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, Tabreed ESG Report 2023/32/ prepared in accordance with the GRI 2021 standards, Federal Decree-Law No. (33) of 2021 /34/, and onsite interviews with relevant personnel /9/ Assessment Conclusion: The assessment team has validated that Tabreed respects internationally recognized human rights through a comprehensive review of its policies and practices. Tabreed adheres to international laws and conventions, including the core conventions of the International

	Labour Organization (ILO) related to forced labour, child labour, and non-discrimination. Specifically, the company complies with: • ILO Conventions No. 29 and No. 105 on the Abolition of Forced Labour, • ILO Conventions No. 138 and No. 182 on the Abolition of Child Labour, and • ILO Conventions No. 100 and No. 111 on Non-discrimination. This commitment has been confirmed through a review of Tabreed's policies outlined on pages 23 and 40 of the ESG report /32/, as well as the company's supplier mandates for effective child labour and human rights policies. The assessment team also conducted interviews with key personnel and the team concluded that Tabreed actively promotes and upholds human rights standards in line with international conventions.
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4.2.7.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk has been identified.	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, project design /10/, authority approvals /11/-/14/, and onsite interviews with relevant personnel /9/ Assessment Conclusion: The project activity has no adverse impact on Indigenous people or cultural heritage. The land for the project was allocated by the Department of Municipalities and Transport, as confirmed through the Land Site Plan (Affection Plan) /11/. Additionally, the assessment team conducted a thorough review using Google Earth imagery to cross-check the site location, ensuring that no communities were residing on the land before the project commenced. Based on this verification, the assessment team concludes that the project activity does not affect indigenous populations or cultural heritage. Additionally, the assessment team has checked and ensured that the project activity has received all necessary approvals /11/-/14/ from the relevant government authorities.

4.2.7.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
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No risk has been identified.	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, project design /10/, authority approvals /11/-/14/, and onsite interviews with relevant personnel /9/ Assessment Conclusion: as described in section 4.2.7.3 above, property rights are protected as confirmed and validated by the Land Site Plan (Affection Plan) /11/ and Google Earth imagery. Additionally, the assessment team has checked and ensured that the project activity has received all necessary approvals /11/-/14/ from the relevant government authorities.
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4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Summary of the benefit sharing plan	There is no benefit sharing is involved; hence not applicable.
Benefit sharing during the monitoring period	There is no benefit sharing is involved; hence not applicable.

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	No risk has been identified. Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, project design /10/, authority approvals /11/-/14/, and onsite interviews with relevant personnel /9/ Assessment Conclusion: As described in section 4.2.7.3 above, the project activity has been constructed on developed land, as confirmed and validated from the Land Site Plan (Affection Plan) /11/. Additionally, Google Earth imagery confirmed that the land was vacant prior to the implementation of the project activity. Therefore, the assessment concludes that there is no adverse impact on biodiversity and ecosystems due to the project.

Soil degradation and soil erosion	The above assessment also concludes that due to the implementation of the project activity, there has been no occurrence of soil degradation or erosion, and no such impacts are expected in the future.
Water consumption and stress	Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, project design /10/, authority approvals /11/-/14/, ADDC invoices /16/ and onsite interviews with relevant personnel /9/ Assessment Conclusion: During the site visit, it was observed that the project activity sources its water from ADDC, which desalinates seawater and provides it to the project. Based on this, the assessment team concludes that the project activity does not place stress on local water resources.

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species or habitat	No risk has been identified. Evidence Gathering Activities: Desk review and onsite assessment Evidence Checked: Registered VCS PD /3/, project design /10/, authority approvals /11/-/14/, Google Earth historical imagery, and onsite interviews with relevant personnel /9/ Assessment Conclusion: The assessment team has validated that the project will not adversely impact habitats for rare, threatened, or endangered species. During the site visit, it was observed that the project is not located in or adjacent to any such habitats. The site location was cross-checked using Google Earth imagery, confirming the absence of critical habitats. Additionally, the Land Site Plan (Affection Plan) /11/ further validated that the land is not in proximity to any protected or sensitive ecological zones. Based on these findings, the assessment team concluded that there is no risk to biodiversity concerning rare, threatened, or endangered species.
Areas needed for habitat connectivity	Not applicable as described above.

Evidence gathering activities, evidence checked, and assessment conclusion	
Habitats for rare, threatened, and endangered species	No risk has been identified as described above
Areas for habitat connectivity	Not applicable as described above.

4.2.8.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
NIL	Not applicable, as the project does not involve the introduction of any species. The activity is focused on the installation of new district cooling plants of various capacities designed to supply cooling services to residential and commercial consumers. This project does not impact ecosystems in terms of species introduction or habitat modification, as the infrastructure is centred on energy-efficient cooling technologies rather than environmental alterations that would involve species relocation or introduction.

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
NIL	As described above, no risk has been identified that could prevent the spread or continued existence of invasive species. The project activity, involving the installation of district cooling plants, does not have an impact on the introduction or spread of invasive species, as it does not involve activities such as land clearing, species introduction, or disruption to habitats that would facilitate the spread of non-native species. The project is focused on infrastructure development with minimal environmental disturbance.

Evidence gathering activities, evidence checked, and assessment conclusion
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Invasive species	No risk of invasive species introduction or spread identified. The project focuses on the installation of district cooling plants, with no activities like land clearing or habitat disruption that would enable the spread of invasive species.
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4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	Not applicable, as the project does not involve any ecosystem conversion. It focuses on the installation of energy-efficient district cooling plants of varying capacities to supply cooling services to residential and commercial consumers. Therefore, there are no significant changes or alterations to natural ecosystems or land use associated with the project.

4.3 Accuracy of Reduction and Removal Calculations

The verification team has confirmed that the equations and choices provided in the methodology, as well as the methodological tools, are correctly quoted in the Monitoring Report (MR) /01/. The emission reductions of the project are calculated using the formulae outlined in the applied methodology AM0117 version 2.0 /6/. The assessment team reviewed the emission reduction spreadsheets (ER sheets) and found the formulae to be accurate and consistent with the monitoring plan of the registered VCS PD and the applied methodology.

Furthermore, the verification team checked the Emission Reduction calculation sheet /2/ and concluded that the equations used have been correctly applied, in line with the selected methodologies AM0117 "Introduction of a new district cooling system" version 2.0 /6/, as well as the registered VCS PD /3/.

Baseline Emissions:

The project proponent has adopted approach 1 as per paragraph 37a) of the applied methodology AM0117 version 2.0 /6/. The baseline emissions are calculated by applying equation 2 of the methodological tool "TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation", version 3.0.

$$BE_{EC,y} = \sum_k EC_{BL,k,y} \times EF_{EF,k,y} \times (1 + TDL_{k,y})$$

Where,

$BE_{EC,y}$ or BE_y = Baseline emissions from electricity consumption in year y (t CO2 / yr)

$EC_{BL,k,y}$ or $Q_{B,y}$ = quantity of electricity that would be consumed by the baseline electricity consumer k in year y (MWh/yr)

$EF_{EF,k,y}$ or $EF_{EL,y}$ = Emission factor for electricity generation for source k in year y (t CO₂/MWh)

$TDL_{k,y}$ or $TDL_{B,y}$ = Average technical transmission and distribution losses for providing electricity to source k in year y

The electricity consumption in the baseline scenario is calculated as per equation 7 of the applied methodology AM0117, version 2.0.

$$Q_{B,y} = \sum_r \frac{OPC_{r,y}}{SEER_{B,i}}$$

Where,

$OPC_{r,y}$ = Cooling output of new district cooling plant r in year y (MWh)

$SEER_{B,i}$ = Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i, defined as energy output divided by energy input (MWh/MWh)

The baseline emissions calculated for both the project activity instances are as below.

Parameter	Description	25-May-2023 to 31-December-2023	01-January-2024 to 31-July-2024	Means of Verification
Project Activity Instance 1 – Sea World DCP, Yas Island (AD-026)				
BE _y	Baseline emissions in year y (tCO _{2e} /yr)	25,477	20,556	Calculated
Q _{B,y}	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year y (MWh/yr)	51,033	41,176	Calculated
OPC _{r,y}	Cooling output of new district cooling plant r in year y (MWh)	115,335	93,057	SAP records/17/, /18/
SEER _{B,y}	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology (MWh/MWh)	2.26	2.26	Registered VCS PD/3/ As validated during the registration stage
TDL _{B,y}	Average technical transmission and distribution losses for providing electricity to the baseline in year y (%)	9.24%	9.24%	Registered VCS PD/3/ As validated from Annual Technical Report 2023/42/

Parameter	Description	25-May-2023 to 31-December-2023	01-January-2024 to 31-July-2024	Means of Verification
EF _{EL,y}	Grid CO2 emission factor (tCO _{2e} /MWh)	0.457	0.457	Registered VCS PD/3/ As validated during the registration stage and ex-ante for the first crediting period
Project Activity Instance 2 –DCP 1 Saadiyat Island (AD-021)				
BE _y	Baseline emissions in year y (tCO _{2e} /yr)	-	8,238	Calculated
Q _{B,y}	Estimated electricity consumption of isolated and less efficient air-cooled reciprocating chiller systems in year y (MWh/yr)	-	16,502	Calculated
OPC _{r,y}	Cooling output of new district cooling plant r in year y (MWh)	-	37,295	SAP records/17/, /18/
SEER _{B,y}	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology (MWh/MWh)	2.26	2.26	Registered VCS PD/3/ As validated during the registration stage
TDL _{B,y}	Average technical transmission and distribution losses for providing electricity to the baseline in year y (%)	9.24%	9.24%	Registered VCS PD/3/ As validated from Annual Technical Report 2023/42/
EF _{EL,y}	Grid CO2 emission factor (tCO _{2e} /MWh)	0.457	0.457	Registered VCS PD/3/ As validated during the registration stage and ex-ante for the first crediting period

As per the registered VCS Project Design (PD) /3/, the baseline emissions are calculated using the lower value of the cooling output measured at both the District Cooling Plant (DCP) and at the customer end. This conservative approach ensures a more cautious estimate of baseline emissions and is therefore accepted by the verification team.

Project Emissions:

The project emissions are calculated by applying equation 11 of the applied methodology AM0117, version 2.0 /6/.

$$PE_y = PE_{EC,y} + PE_{FC,j,y}$$

Where,

PE_y = Project emissions in year y (tCO_{2e})

$PE_{EC,y}$ = Emissions from electricity consumption associated with the generation of cooling output in the new district cooling plant(s) (tCO_{2e})

$PE_{FC,j,y}$ = CO₂ emissions from fossil fuel combustion associated with the generation of cooling output in the new district cooling plant(s) (tCO_{2e})

During the current monitoring period, the project has not consumed fossil fuels for cooling production, except for minor testing of backup generators. Since the fuel consumption for this purpose is negligible, it has been excluded from the emissions calculations. Consequently, CO₂ emissions associated with fossil fuel combustion for the generation of cooling output in the new district cooling plant do not apply.

The project emissions from the consumption of electricity by the project activity ($PE_{EC,y}$) are calculated using equation 2 (explained above in baseline emissions) of the "TOOL05: Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation.", version 3.0.

$$PE_{EC,j,y} = EC_{PJ,j,y} \times EF_{EL,y} \times (1 + TDL_y)$$

Where,

$PE_{EC,y}$ = Project emissions from electricity consumption associated with the generation of cooling output in the district cooling plant in year y (t CO₂ / yr)

$EC_{PJ,y}$ = quantity of electricity consumed by the new district cooling plant j in year y (MWh/yr)

$EF_{EL,y}$ = Grid CO₂ emission factor (tCO_{2e}/MWh)

TDL_y = Average technical transmission and distribution losses for providing electricity in year y (%)

Parameter	Description	25-May-2023 to 31-December-2023	01-January-2024 to 31-July-2024	Means of Verification
Project Activity Instance 1 – Sea World DCP, Yas Island (AD-026)				

Parameter	Description	25-May-2023 to 31- December- 2023	01-January- 2024 to 31- July-2024	Means of Verification
$PE_{EC,y}$	Project emissions from electricity consumption associated with the generation of cooling output in the district cooling plant in year y (t CO ₂ / yr)	14,442	11,019	Calculated
$EC_{PJ,y}$	Quantity of electricity consumed by the new district cooling plant j in year y (MWh/yr)	28,927	22,070	SAP Records /17/ ADDC Invoices /16/
TDL_y	Average technical transmission and distribution losses for providing electricity to the baseline in year y (%)	9.24%	9.24%	Registered VCS PD /3/ As validated from Annual Technical Report 2023/42/
$EF_{EL,y}$	Grid CO ₂ emission factor (tCO _{2e} /MWh)	0.457	0.457	Registered VCS PD /3/ As validated during the registration stage and ex-ante for the first crediting period
Project Activity Instance 2 –DCP 1 Saadiyat Island (AD-021)				
$PE_{EC,y}$	Project emissions from electricity consumption associated with the generation of cooling output in the district cooling plant in year y (t CO ₂ / yr)	-	5,937	Calculated
$EC_{PJ,y}$	Quantity of electricity consumed by the new district cooling plant j in year y (MWh/yr)	-	11,892	SAP Records /17/ ADDC Invoices /16/
TDL_y	Average technical transmission and distribution losses for providing electricity to the baseline in year y (%)	9.24%	9.24%	Registered VCS PD /3/ As validated from Annual Technical Report 2023/42/

Parameter	Description	25-May-2023 to 31-December-2023	01-January-2024 to 31-July-2024	Means of Verification
EF _{EL,y}	Grid CO2 emission factor (tCO _{2e} /MWh)	0.457	0.457	Registered VCS PD /3/ As validated during the registration stage and ex-ante for the first crediting period

As per the registered VCS Project Design (PD) /3/, the project emissions are calculated using the higher value of electricity consumption measured by Tabreed's internal meters and from utility invoices (ADDC). The verification team has reviewed this method and found it to be conservative, appropriate, and acceptable, ensuring that the calculations align with the registered monitoring plan and applied methodology.

Leakage Emissions:

The leakage emissions are calculated by applying equation 12 of the applied methodology AM0117, version 2.0 /6/

$$LE_y = LE_{Ref,y} + LE_{Ref,sc} + LE_{water}$$

Where,

LE_y = Leakage emissions in year y (tCO_{2e})

$LE_{Ref,y}$ = Refrigerant leakage emissions from the project in year y (tCO_{2e})

$LE_{Ref,sc}$ = Refrigerant leakage emissions from the baseline cooling equipment that is scrapped as a result of the project activity (only accounted in the first year of the crediting period) (tCO_{2e})

$LE_{water,y}$ = Emissions due to the freshwater usage in the project system in year y (tCO_{2e})

During the site visit, the assessment team reviewed the maintenance records for the chiller and verified that no refrigerants were used during the current monitoring period. Additionally, the verification team confirms that refrigerant leakage emissions from baseline cooling equipment are not applicable to the project activity. The following observations support this conclusion:

New Construction for Project Activity Instances:

- The District Cooling Plants (DCPs) constructed as part of the project activity are associated with new buildings or facilities. For Project Activity PA01, the DCP was specifically built for the

SeaWorld facility, a newly established customer facility designed explicitly for district cooling utilization.

Integration with Existing Infrastructure for PA02:

- Although a few of customers facilities for PA02 were pre-existing, they were previously serviced by other DCPs. The new DCP for PA02 has been integrated into the existing chilled water network to meeting increasing demand of cooling.

Consequently, both $LE_{Ref,y}$ and $LE_{Ref,sc}$ values are zero for the current monitoring period.

According to the registered VCS Project Description (PD) /3/, the emissions linked to make-up water consumption are considered negligible. Therefore, the leakage emissions from water consumption are equivalent to the basin volume of cooling towers and accounted for only once, during the first monitoring period, using the applicable formula and approach outlined in equation 15 of the methodology.

$$LE_{water} = Q_{water} \times EF_{water}$$

Where,

LE_{water} = Emissions due to the freshwater usage in the project system (tCO_{2e})

Q_{water} = Maximum designed quantity of fresh water to be used in the project system (t)

EF_{water} = Emission factor associated with the production of freshwater (tCO_{2e}/t)

Parameter	Description	Sea World DCP, AD-026	Saadiyat Island DCP 1, AD-021	Means of Verification
LE_{water}	Emissions due to the freshwater usage in the project system (tCO _{2e})	19	26	Calculated
Q_{water}	Maximum designed quantity of fresh water to be used in the project system (t)	1620.5	2,260	AD026 – Design Report /35/. As validated during the registration stage AD021 – Design Report /10/
EF_{water}	Emission factor associated with the production of freshwater (kgCO _{2e} /t)	11.5	11.5	Registered VCS PD /3/ As validated during the registration stage

Accordingly, the emission reductions are calculated using equation 17 of the applied methodology AM0117 version 2.0 /6/.

$$ER_y = BE_y - PE_y - LE_y$$

Therefore, $ER_y = 54,271 - 31,398 - 45$

$$ER_y = 22,828 \text{ tCO}_2\text{e}$$

The verification team confirms that all parameters used in the calculations are accurate, verifiable, and transparent, with assumptions clearly outlined and based on evidence. Calculations align with the pre-defined formulae in the registered VCS PD /3/. Emission reductions were calculated in line with the applied methodology AM0117, version 2.0 /6/, and the registered VCS PDD /3/. The Project Proponent (PP) utilized both monitored data and ex-ante fixed data, including default values as allowed by the methodology.

The verification team has thoroughly reviewed the values used for calculating GHG emission reductions and found them accurate and appropriate. The following parameters are determined ex-ante and detailed in the VCS PD:

Monitoring Parameters determined ex-ante (available at validation)

Parameter	Description	Value	Assessment
Building Type	The building types are categorized as new. The project will be providing cooling to new buildings.	-	The included project activity instances are new district cooling plants (buildings) as checked and verified from Detailed design report /10/, Affection Plan /11/, and regulatory approvals /12/-/14/.
Baseline cooling technologies	The data collection has been done by a third party based on various data sources, including Abu Dhabi Department of Energy (DoE), Abu Dhabi Industrial Strategy (ADIS), Abu Dhabi Department of Economic Development (ADDED), Abu Dhabi Industrial Development Bureau (IDB) and Global Market Insights (third party consultant).	*Values included below this table	Abu Dhabi Cooling Market Report 2023-2032 /20/.

Parameter	Description	Value	Assessment
EF _{grid,CM,y}	Combined margin emission factor for the grid (tCO _{2e} /MWh)	0.457	IRENA UAE Energy profile published 08-August-2023 The assessment team further compared the emission factor with Harmonized IFI Default Grid Factors 2021 v3.2 from UNFCCC /37/ and the IRENA emission factor was found conservative. The emission factor is ex-ante for the first crediting period.
Q _{water}	Maximum designed quantity of fresh water to be used in the project system (Ton)	1620.52 – AD026 2,260 – AD021	AD026 – Design Report /35/. As validated during the registration stage AD021 – Design Report /10/

*Baseline cooling technologies:

Technology	SEER	Cooling Output in Million TR	Cumulative Cooling Output (million TR)
District Cooling	4.84	0.63	0.63
Split Units	4.51	0.58	1.21
Centralized Water Cooled	4.31	0.27	1.48
Others (Ducted & Packaged Cooling Systems)	3.52	1.03	2.51
Air-cooled alone	2.26	1.04	3.55
Total		3.55	

The spreadsheet submitted by the Project Proponent transparently presents all data parameter values used in the emission reduction calculations. The input values have been verified against reliable sources, including monitoring records, MR /1/, and the applied methodology /6/. Emission reductions were cross-checked with the emission reduction spreadsheet /2/ and confirmed as accurate. No significant reporting risks were identified in the reported data.

Details of the monitoring parameters used for emission reduction calculations are provided below:

Data and Parameters Monitored

Monitoring Parameter Requirement	Assessment/ Observation by the VVB		
Data / Parameter: (as in the monitoring plan of VCS PD):	Cooling output of new district cooling plant r in year y ($OPC_{r,y}$) Unit - TRh (It is converted into MWh to compare SEER against the baseline)		
Measuring frequency/Time Interval:	Continuous monitoring and daily recording		
Reported value:	PA	25-May-2023 to 31-December-2023 (MWh _{cooling})	01-January-2024 to 31-July-2024 (MWh _{cooling})
	PA01	115,335	93,057
	PA 02	0	37,295
	Above reported values are error adjusted values.		
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes		
Details of monitoring equipment:	BTU Meters both at DCP Plants and Customer end		
	Particulars	PA-01 (AD026)	PA-02 (AD021)
	Instrument Name	BTU Meter	BTU Meter
	Manufacturer	Endress+ Hauser	Endress+ Hauser Kamstrup (customer meters)
	Serial Numbers/ Meter ID	NTP01BT1 (DCP) CUC01BT000 8000317 (Customer)	NTP01BT1 (DCP) Customer Meters: CUC02BT000 8000228 CUC01BT000 8000234 CUC11BT000 8000227

Monitoring Parameter Requirement	Assessment/ Observation by the VVB		
			CUC14BT000 8000235 CUC17BT000 8000235 CUC01BT000 8000519
	Accuracy Class	0.2	0.2
	Since the calibration certificates (/22/, /23/) do not cover the entire monitoring period, the project proponent has conservatively applied an error equivalent to the meter's accuracy class (0.2%) to the entire monitoring period. The assessment team has reviewed this approach and found it to be conservative; therefore, it is accepted.		
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Registered VCS PD doesn't specify the accuracy of the monitoring equipment. However, the accuracy class of the BTU meter is observed to be 0.2 for both BTU meters at DCP plants and at the customer end.		
Calibration frequency /interval: Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	As per registered PD, calibration frequency is annual.		
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	The annual calibration frequency specified in the registered VCS PD (/3/) was not followed, and a gap in the calibration of the meters was observed (/22/, /23/). To justify this deviation, the client provided the <i>Metering and Data Exchange Code</i> issued by the Department of Energy, Abu Dhabi (/39/), which stipulates that water flow meters require on-site verification every two years and calibration every five years. While this differs from the annual calibration frequency outlined in the registered VCS PD, the project proponent has		

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
	applied the maximum permissible error for the entire monitoring period. The assessment team considers this a suitably conservative approach in line with para 4a) of Guidelines for assessing compliance with the calibration frequency, version 01/40/ and, therefore, accepts it.
The company performing the calibration (internal or external calibration):	External
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Yes
Is (are) calibration(s) valid for the whole reporting period?	No
If applicable, has the reported data been cross-checked with other available data?	The cooling output data generated at District Cooling Plant (DCP) sites is consistently compared with cooling energy consumption data measured at the Energy Transfer Stations (ETS). For baseline emissions calculations, the lower of the two values is conservatively selected for each project instance to ensure accuracy and alignment with methodology requirements. This approach is consistent with the registered VCS PD /3/.
How were the values in the monitoring report verified?	The reported data is checked and verified from the SAP records /17/ at DCP plants and SAP records /18/ at the customer end.
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures the correct transfer of data and reporting of emission reductions, and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB		
Data / Parameter: (as in monitoring plan of VCS PD):	Quantity of electricity consumed by the new district cooling plant j in year y ($EC_{PJ,j,y}$) Unit - MWh		
Measuring frequency/Time Interval:	Continuous monitoring and daily recording – PMU (Internal Meters) Continuous monitoring and monthly recording – Utility (ADDC) meters		
Reported value:	PA	25-May-2023 to 31-December-2023 (MWh)	01-January-2024 to 31-July-2024 (MWh)
	PA01	28,927	22,070
	PA 02	0	11,892
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes		
Details of monitoring equipment:	Energy Meters		
	Particulars	PA-01 (AD026)	PA-02 (AD021)
	Instrument Name	Energy Meter	Energy Meter
	Manufacturer	Elster (ADDC meters)	Elster (ADDC meters)
	Serial Numbers/ Meter ID	EIM0001K, EIM0002K ADDC Meters E2T12D000 237, E2T12D000 238	EIM0001K, EIM0002K, EIM0003K, EIM0005K ADDC Meters E2T12D000 153, E2T12D000 154, E2T12D000 155, I1C14D005 987

Monitoring Parameter Requirement	Assessment/ Observation by the VVB		
	Accuracy Class	0.2	0.2
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Registered VCS PD doesn't specify the accuracy of the monitoring equipment. However, the accuracy class of the energy meters is observed to be 0.2 for both ADDC and PMU meters.		
Calibration frequency /interval: Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	As per registered PD, calibration frequency is annual.		
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	The annual calibration frequency for the internal PMU meters, as specified in the registered VCS PD (/3/), was not adhered to, and gaps in meter calibration were observed (/22/, /23/). To justify this deviation, the client provided the <i>Metering and Data Exchange Code</i> issued by the Department of Energy, Abu Dhabi (/39/), which requires energy meters to undergo on-site verification every two years and calibration every five years. While this differs from the annual calibration frequency specified in the VCS PD, the project proponent has conservatively applied the maximum permissible error for the entire monitoring period in line with para 4a) of Guidelines for assessing compliance with the calibration frequency, version 01/40/. Furthermore, since the higher value of energy consumption, as recorded by either the internal PMU meters or the ADDC meter, is used for the emission reduction (ER) calculations, applying the maximum permissible error does not impact the ER calculations. The assessment team finds this approach conservative and, therefore, accepts it.		
The company performing the calibration (internal or external calibration):	External		

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
	Since utility meters are in the custody of ADDC, calibration by the project proponent is not applicable.
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Yes
Is (are) calibration(s) valid for the whole reporting period?	No
If applicable, has the reported data been cross-checked with other available data?	The electricity consumption data measured at internal PMU meters is consistently compared to electricity consumption as per ADDC invoices. For project emissions calculations, the higher of the two values is conservatively selected for each project instance to ensure accuracy and alignment with methodology requirements. This approach is consistent with the registered VCS PD/3/.
How were the values in the monitoring report verified?	The reported data is checked and verified from the SAP records/17/ for PMU meters and ADDC invoices/16/.
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures the correct transfer of data and reporting of emission reductions, and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in the monitoring plan of VCS PD):	Average technical transmission and distribution losses for providing electricity in year y ($TDL_{B,y}$, and TDL_y) Unit - %

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Measuring frequency/Time Interval:	Annual In the absence of annual data updates, the latest values can be used, but not older than 5 years.
Reported value:	Transmission loss: 2.04% Distribution loss: 7.2%
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes The latest available values of transmission and distribution losses from Annual Technical Report 2023/42/, as these figures are not older than five years.
Details of monitoring equipment:	Not Applicable
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Not Applicable
Calibration frequency /interval: Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	Not Applicable
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	Not Applicable
The company performing the calibration (internal or external calibration):	Not Applicable
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Not Applicable
Is (are) calibration(s) valid for the whole reporting period?	Not Applicable
If applicable, has the reported data been cross-checked with other available data?	Not Applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
How were the values in the monitoring report verified?	The values validated at the time of registration of the project are applied.
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Publicly available data is applied.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in the monitoring plan of VCS PD):	Benchmark Seasonal Energy Efficiency Ratio of the baseline cooling technology i (SEER _{B,i}) Unit – MWh/MWh
Measuring frequency/Time Interval:	Every 3 years
Reported value:	2.26
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes The benchmark SEER determined at the time of validation still applies as it is within 3 years of monitoring frequency.
Details of monitoring equipment:	Not Applicable
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Not Applicable
Calibration frequency /interval:	Not Applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	Not Applicable
The company performing the calibration (internal or external calibration):	Not Applicable
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Not Applicable
Is (are) calibration(s) valid for the whole reporting period?	Not Applicable
If applicable, has the reported data been cross-checked with other available data?	Not Applicable
How were the values in the monitoring report verified?	The values validated at the time of registration of the project are applied. It is further checked and verified from the baseline study report/20/.
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	It is determined from third-party Abu Dhabi Cooling Market Report 2023-2032 prepared by Global Market Insights (GMI), a study conducted in 2023/20/. No QA/QC at the project site applies.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in the monitoring plan of VCS PD):	Quantity of refrigerant k used in the project in year y ($R_{k,y}$)

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
	Unit – Ton
Measuring frequency/Time Interval:	Continuous, on a consumption basis
Reported value:	PA01 – Zero PA02 - Zero
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Not Applicable
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Not Applicable
Calibration frequency /interval: Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	Not Applicable
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	Not Applicable
The company performing the calibration (internal or external calibration):	Not Applicable
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Not Applicable
Is (are) calibration(s) valid for the whole reporting period?	Not Applicable
If applicable, has the reported data been cross-checked with other available data?	Not Applicable
How were the values in the monitoring report verified?	During the site visit, the assessment team reviewed the maintenance records for the chillers and

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
	verified that no refrigerant was consumed throughout the current monitoring period.
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data management ensures the correct transfer of data and reporting of emission reductions, and all necessary QA/QC processes are in place.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in the monitoring plan of VCS PD):	Global Warming Potential of the refrigerant k (R134A) (GWPk,) Unit – NIL
Measuring frequency/Time Interval:	As and when new IPCC reports are published with updated global warming potential values.
Reported value:	1,530
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Details of monitoring equipment:	Not Applicable
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Not Applicable
Calibration frequency /interval: Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	Not Applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	Not Applicable
The company performing the calibration (internal or external calibration):	Not Applicable
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Not Applicable
Is (are) calibration(s) valid for the whole reporting period?	Not Applicable
If applicable, has the reported data been cross-checked with other available data?	Not Applicable
How were the values in the monitoring report verified?	The reported value of GWP of R134a is checked and verified from IPCC AR6 https://ghgprotocol.org/sites/default/files/2024-08/Global-Warming-Potential-Values%20%28August%202024%29.pdf
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Publicly available data is applied.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Data / Parameter: (as in the monitoring plan of VCS PD):	Emission factor associated with the production of freshwater (EF _{water}) Unit – tCO _{2e} /t

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
Measuring frequency/Time Interval:	First Monitoring Period (as per registered VCS PD)
Reported value:	0.0115
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes The PP applied the value validated at the time of registration of the project.
Details of monitoring equipment:	Not Applicable
Is the accuracy of the monitoring equipment as stated in the VCS PD? If the PD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Not Applicable
Calibration frequency /interval: Is it as per monitoring methodology /CDM EB guidance / local or national standards/manufacturers specification	Not Applicable
Is the calibration interval in line with the monitoring plan of the VCS PD? If the PD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	Not Applicable
The company performing the calibration (internal or external calibration):	Not Applicable
Did calibration confirm the proper functioning of monitoring equipment? (Yes / No):	Not Applicable
Is (are) calibration(s) valid for the whole reporting period?	Not Applicable
If applicable, has the reported data been cross-checked with other available data?	Not Applicable
How were the values in the monitoring report verified?	PP applied the value of the emission factor of desalination same as validated at the time of project registration.

Monitoring Parameter Requirement	Assessment/ Observation by the VVB
	The assessment team further evaluated the selected values for appropriateness and conservativeness by cross-referencing them with data from EWEK statistical reports/38/. The applied values were verified to be conservative, ensuring that emission reduction calculations as conservative.
Does the data management (from data generation to emission reduction calculation) ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Publicly available data is applied.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied, or has a request for deviation been approved?	Not applicable

4.4 Quality of Evidence to Determine Reductions and Removals

When verifying the emission reductions, DNVBAPL ensured a comprehensive audit trail was in place, containing all evidence and records necessary to validate the figures presented in the report. All source documents, which underpin the assumptions and other GHG data, are referenced and detailed above.

In the review of the audit trail, DNVBAPL examined the following aspects:

- Sufficiency of Evidence:** The team ensured that evidence was available frequently throughout the monitoring period and covered the full duration.
- Source and Nature of Evidence:** The team assessed the authenticity and reliability of the evidence provided.
- Cross-Verification with Other Sources:** Where applicable, DNVBAPL cross-checked the monitoring report against other available sources of comparable data to confirm the accuracy of the reported figures. All sources referenced are included in Appendix 2.

The verification team confirms that the monitoring equipment used for the project activity adhered to the requirements outlined in the registered monitoring plan. All BTU meters and energy meters have been calibrated ensuring measurement accuracy, though the specified frequencies in the registered VCS PD/3/ wasn't followed. Since, the project proponent has applied the maximum permissible error equal to meter

accuracy class conservatively, the assessment team has accepted the conservative approach. Calibration was done by a certified professional as per the VCS PD guidelines, with records reviewed and found satisfactory during the site visit.

DNVBAPL also confirmed that the data collection system adhered to the monitoring plan requirements, ensuring that data acquisition, aggregation, and management practices were aligned with best practices. The monitoring personnel were well-trained, following reproducible routines that ensured accuracy and competence in carrying out the relevant tasks.

4.5 Non-Permanence Risk Analysis

The project “TABREED DISTRICT COOLING GROUP PROJECT” (VCS 4519) does not fall under the category of the AFOLU project. Hence, non-permanence risk does not apply to this project activity.

5 VERIFICATION OPINION

5.1 Verification Summary

DNV Business Assurance India Private Limited (hereinafter referred to as DNVBAPL) hereby declares that it has conducted the first verification of the renewable (first) crediting period for the monitoring period from 25-May-2023 to 31-July-2024, inclusive, for the project activity titled “TABREED DISTRICT COOLING GROUP PROJECT” under VCS Registry Project ID 4519. This verification was performed in accordance with the applicable requirements for VCS activities. The project proponent has reported a total of 22,828 tCO_{2e} in emission reductions for this period, assessed against the criteria outlined in VCS Standard, version 4.7 /5/.

The project proponent IS responsible for:

- Preparing greenhouse gas (GHG) emissions data and reporting on GHG emission reductions as per the monitoring plan specified in the registered VCS PD/3/ and Monitoring Report /1/.
- Developing and maintaining records and reporting procedures aligned with the monitoring plan, including accurate calculation and determination of the project’s GHG emission reductions.

DNVBAPL's responsibility is to provide an independent verification opinion of the project's compliance with VCS Standard version 4.7/5/ and the applied GHG program, specifically concerning the reported GHG emission reductions from the project.

Based on documented evidence and verification through an on-site assessment, DNVBAPL confirms the following:

- The project has been implemented and operated in alignment with the registered project description and monitoring report, version 1.6, dated 14-April-2025/1/.
- The monitoring report, along with all supporting documentation, is comprehensive, verifiable, and compliant with the requirements of VCS Standard, version 4.7/5/.
- The monitoring plan has been effectively implemented in accordance with the approved baseline and monitoring methodology /6/ and as outlined in the registered Project Design and Monitoring Report (MR) /1/.

DNVBAPL, accredited by an International Accreditation Forum-recognized accreditation body, has verified the GHG emissions statement for the “TABREED DISTRICT COOLING GROUP PROJECT” in compliance with ISO 14064-3, with the applicable version, ISO 14064-3:2019.

5.2 Verification Conclusion

DNVBAPL's verification approach is designed around assessing the risks associated with the accurate reporting of GHG emission data and the existing controls to mitigate those risks. DNVBAPL planned and

conducted the verification by gathering relevant evidence, information, and clarifications, to provide a Reasonable Assurance that the reported GHG emission reductions are fairly presented.

Based on the verified monitoring report, the emission reduction (ER) calculation spreadsheet, and findings from both the on-site audit and desk review, the verification team confirmed that both the project activity instances included in the grouped project activity are operational during the current monitoring period. Additionally, it was verified that the grouped project only reports reductions in GHG emissions, amounting to an emission reduction of **22,828 tCO_{2e}** for this monitoring period.

Verification period: From 25-May-2023 to 31-July-2024

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO _{2e})	Project emissions (tCO _{2e})	Leakage emissions (tCO _{2e})	Reduction VCU (tCO _{2e})	Removal VCU (tCO _{2e})	Total VCUs (tCO _{2e})
Project Activity Instance 01 – Sea World DCP (AD-26)						
25-May-2023 to 31-December-2023	25,477	14,442	19	11,016	-	11,016
01-January-2024 to 31-July-2024	20,356	11,019	0	9,537	-	9,537
Project Activity Instance 02– Saadiyat Island DCP 1 (AD-21)						
25-May-2023 to 31-December-2023	0	0	0	0	-	0
01-January-2024 to 31-July-2024	8,238	5,937	26	2,275	-	2,252

Vintage period	Baseline emissions (tCO _{2e})	Project emissions (tCO _{2e})	Leakage emissions (tCO _{2e})	Reduction VCU (tCO _{2e})	Removal VCU (tCO _{2e})	Total VCUs (tCO _{2e})
25-May-2023 to 31-December-2023	25,477	14,442	19	11,016	-	11,016
01-January-2024 to 31-July-2024	28,794	16,956	26	11,812	-	11,812
Total	54,271	31,398	45	22,828	-	22,828

5.3 Ex-ante vs Ex-post ERR Comparison

Project Activity Instance 01 – Sea World DCP (AD-26)

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
25-May-2023 to 31-December-2023	11,681	11,016	-5.69%	Lesser cooling output as compared to the estimated
01-January-2024 to 31-July-2024	11,272	9,537	-15.39%	Lesser cooling output as compared to the estimated As per registered VCS PD, ex-ante ERs for year 2024 are 19,323 tCO _{2e} . The ex-ante ERs are determined by proportionally allocating the total based on a 7-month monitoring period within the year.
Total	22,953	20,553	-10.46%	

Project Activity Instance 02 – Saadiyat Island DCP 1 (AD-21)

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
25-May-2023 to 31-December-2023	-	-	-	PA-02 commissioned on 03-May-2024/7/
01-January-2024 to 31-July-2024	4,646	2,275	-51.03%	Lesser cooling output as compared to the estimated As per registered VCS PD, ex-ante ERs for year 2024 are 19,323 tCO _{2e} . The ex-ante ERs are determined by proportionally allocating the total based on an 88-day monitoring period since the commissioning of PA-2.
Total	4,646	2,275	-51.03%	

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Not applicable as there is no commercial sensitive information included in the monitoring report as well as this verification report.

APPENDIX 2: REFERENCE DOCUMENTS

Ref	Documents
/1/	- Monitoring Report version 01, dated 17-September-2024 - Monitoring Report version 1.1, dated 02-November-2024 - Monitoring Report version 1.2, dated 07-November-2024 - Monitoring Report version 1.3, dated 16-November-2024 - Monitoring Report version 1.4, dated 04-December-2024 - Monitoring Report version 1.5, dated 10-April-2025 - Monitoring Report version 1.6, dated 14-April-2025
/2/	- ER Sheet corresponding to Monitoring Report version 01, dated 17-September-2024 - ER Sheet corresponding to Monitoring Report version 1.1, dated 02-November-2024 - ER Sheet corresponding to Monitoring Report version 1.2, dated 07-November-2024 - ER Sheet corresponding to Monitoring Report version 1.3, dated 16-November-2024 - ER Sheet corresponding to Monitoring Report version 1.4, dated 04-December-2024 - ER Sheet corresponding to Monitoring Report version 1.5, dated 10-April-2025 - ER Sheet corresponding to Monitoring Report version 1.6, dated 14-April-2025
/3/	Registered PCS PD version 01.9 dated 10-May-2024 https://registry.terra.org/app/projectDetail/VCS/4519
/4/	Validation Report version 1.2 dated 20-May-2024 https://registry.terra.org/app/projectDetail/VCS/4519
/5/	VCS Requirements - VCS Standard (v4.7) - VCS Program Guide (v4.4) - VCS Program Definitions version (v4.5) - Registration & Issuance Process (v4.6) - VCS Validation and Verification Manual version (v3.2) - VCS MR template version 4.4
/6/	Applied Methodology, AM0117: Introduction of a new district cooling system, version 2.0 https://cdm.unfccc.int/methodologies/DB/EOEGREXNGR6VOFAP0PR7DOERUS000X
/7/	Commissioning Certificates - Testing and Commissioning (T&C) Report from Johnson Controls (JCI Project# P211153540) dated 25-May-2023 - Midea Centrifugal Chiller Commissioning Report from Midea (Chiller 3) dated 03-05-2024 (part of the service manual)
/8/	DNVBAPL's Process Requirement Procedures (ICP-2-CCS-PRP), version 6.1
/9/	Site visit attendance sheet dated 15-October-2024 and 16-October-2024
/10/	Detailed Design Report DCP 1 – Option 4 Abu Dhabi, dated 27-July-2022
/11/	Land Site Plan (Affection Plan) from Department of Municipalities and Transport, Abu Dhabi City Municipality – Saadiyat Island DCP
/12/	NOC Report for the plot from Department of Municipalities and Transport, Abu Dhabi City Municipality dated 14-April-2022 – Saadiyat Island DCP

Ref	Documents
/13/	Building Permit from Department of Municipalities and Transport, Abu Dhabi City Municipality – Saadiyat Island DCP Permit Number: B1S-2022-002866-P01
/14/	Saadiyat Cooling Services License (Document No. ED/L01/112 Issue No. 1 Rev (0), Issue Date 14-March-2021) from Department of Energy – Saadiyat Island DCP
/15/	Commissioning Report from Midea for Project DCP -1 Tabreed dated 03-May-2024– Saadiyat Island DCP
/16/	Saadiyat Island DCP 1 - ADDC Invoices for the months of June, July, and August 2024 Sea World DCP – ADDC Invoices for the months from June 2023 – August 2024
/17/	SAP (daily) records for cooling output and electricity consumption (through internal meters) for both the project activity instances
/18/	SAP (daily) records for cooling energy consumption (through BTU meters) at the customer’s end for both the project activity instances
/19/	Month cooling generation and electricity consumption reports generated from the SAP system for both the project activity instances
/20/	Abu Dhabi Cooling Market Report 2023-2032 prepared by Global Market Insights (GMI); study conducted in 2023
/21/	Acknowledgment report for BTU meter change in AD21 dated 01-June-2024
/22/	Calibration Certificates of BTU meters for Plant AD26 - Factory Calibration Certificate for BTU meter (Sr. No. T4045F043BC and T40460043BC) dated 21-April-2022 - Verification report for flow meter (Sr. No. T4209C1900D) dated 04-June-2024 at DCP - Verification report for supply temperature gauge and return temperature gauge (for BTU meter Sr. No. T4045F043BC) dated 04-June-2024 at DCP - Verification report for flow meter (Sr. No. R929BA1900D) dated 04-June-2024 at SeaWorld - Verification report for supply temperature gauge and return temperature gauge (for BTU meter Sr. No. R90472043BC) dated 04-June-2024 at SeaWorld
/23/	Calibration Certificates BTU meters for Plant AD21 - Factory Calibration Certificate for BTU meter (Sr. No. V600E2043BC and V600E1043BC) dated 03-July-2023 Calibration Certificates energy meters for Plant AD21 - Meter test report for PMU meter at Incomer 1 (Sr. MZ-2302A044-01) dated 20-February-2024 - Meter test report for PMU meter at Incomer 2 (Sr. MZ-2302A042-01) dated 20-February-2024 - Meter test report for PMU meter at Incomer 1 (Sr. MZ-2302A053-01) dated 05-March-2024
/24/	Declaration Letter from Tabreed (Ref: TPM-LT-DC 2023-XX-082) dated 31-October-2023 confirming. - The project was neither registered nor rejected in any other GHG programme other than VERRA.

Ref	Documents
	- The project is a voluntary activity that is not covered under any national/local regulations in UAE. - The project has not obtained any ODA funding for the project activity - The project will not claim any GHG benefits in other GHG schemes other than VERRA.
/25/	Stakeholder ongoing communication evidence - Customer Service Activity Reports from May 2023 to July 2024 for both the project activity instances i.e. AD21 and AD26 - Tabreed Enquiry Process v1.0
/26/	Authority approvals including environmental approvals from EAD (B1N-2019-010544) - NOC Report - B1S-2022-002866
/27/	ISO 14064-3:2019 Greenhouse gases Part 3: Specification with guidance for the verification and validation of greenhouse gas statements
/28/	Local Stakeholder Consultation Documents 1. Invitation email dated 06-September-2023 including the list of attendees 2. Screenshots of local stakeholder meeting conducted via MS Teams dated 19-September-2023 3. Presentation content (copy) of the local stakeholder meeting
/29/	Benefits of district cooling in the Context Middle East region https://districtcooling.pro/what-are-the-benefits-of-district-cooling/#Economic_Benefits_to_the_Nation
/30/	Customer Service Charter 1. https://www.tabreed.ae/customer/ 2. https://www.tabreed.ae/wp-content/uploads/2023/04/Customer-Service-Charter-English.pdf
/31/	About Tabreed, company profile, projects https://www.tabreed.ae/about-us/
/32/	Tabreed ESG Report 2023 https://www.tabreed.ae/wp-content/uploads/2024/06/Tabreed-ESG-Report-2023.pdf
/33/	National Central Cooling Company PJSC - Tabreed, Investor Presentation https://www.tabreed.ae/wp-content/uploads/2020/01/Tabreed-Investor-Presentation-Nov-2019-v1.pdf
/34/	Federal Decree-Law No. (33) of 2021
/35/	Concept Design Report (Report No. P131-RP-M-08 R00) dated 15-January-2021 by Fitton Engineering Consultancy LLC
/36/	IRENA UAE Energy profile published 08-August-2023 Accessible Report https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle-East/United-Arab-Emirates_Middle-East_RE_SP.pdf dated 31-July-2023
/37/	Harmonized IFI Default Grid Factors 2021 v3.2 https://unfccc.int/documents/461676
/38/	EWEC Statistical Reports https://www.ewec.ae/en/media/statistical-reports

Ref	Documents
/39/	The Metering and Data Exchange Code (Document No. CD/C01/002) dated Dec 2021 version 5.0 issued by Department of Energy, Abu Dhabi
/40/	Guidelines for assessing compliance with the calibration frequency, version 01 https://cdm.unfccc.int/Reference/Guidclarif/iss/iss_guid05.pdf
/41/	Approved exemption from VERRA “4519_VCS_EXEMPTION_Approved” dated 08-April-2025
/42/	Annual Technical Report 2023 from Department of Energy (DOE), Abu Dhabi available at https://doe.gov.ae/-/media/Project/DOE/Department-Of-Energy/Media-Center-Publications/technical-reporten.pdf (last accessed on 17-April- 2025)

APPENDIX 3: FINDINGS OVERVIEW

Findings Overview Summary

	CARs	CLs	FARs
Total Number raised	02	08	01

Date:	17-October-2024		Raised by:	Vikas Bankar	
Type:	CL	Number:	#1	Reference:	Title Page, Section 1.2 of MR, v1,
Lead Assessor Comment:					
1. Original Date of Issue and VCS Standard Version Please verify the accuracy of the original date of issue and the VCS standard version cited on the title page of MR v1. 2. VVB Name and Verification Period in Section 1.2 Kindly confirm the correctness of the VVB name listed in section 1.2 of MR v1. Additionally, we request verification of the number of years referenced for the project’s verification.					
Client’s Response :				Date : 02-November-2024	
1. The MR has been revised to incorporate the correct date i.e.17-September-2024. 2. VVB name is corrected as per the name available on the VCS website					
Documentation Provided by the client:					
1. Revised MR version 1.1 dated 02-November-2024					
Information Verified by Lead Assessor:					
Revised monitoring report version 1.1 dated 02-November-2024					
The reasoning for not Acceptance or Acceptance and Close Out:				Date: 04-November-2024	
The revised Monitoring Report (MR) version 1.1 dated 02-November-2024, has been updated to include the original date of issue and the corrected version of the VCS standard on the title page, which is deemed appropriate and therefore accepted. Additionally, the Project Proponent (PP) has accurately stated the name of the Validation and Verification Body (VVB) and the duration of the verification period in Section 1.2 of the revised MR, which is also accepted.					
CL#1 is CLOSED.					

Date:	17-October-2024		Raised by:	Vikas Bankar	
Type:	CL	Number:	#2	Reference:	Section 1.8 of MR, v1,
Lead Assessor Comment:					
Inconsistent Format for Geo-Coordinates Please clarify why a consistent format for the geo-coordinates has not been followed across both project activity instances. Longitude for Project Activity Instance 2 Kindly verify the correctness of the longitude mentioned for Project Activity Instance 2.					

Client's Response :	Date : 02-November-2024
1. The geo-coordinates for PA02 have been corrected in the consistent format with PA01 and MR is revised. 2. The geo-coordinates for PA02 have been corrected in the revised MR.	
Documentation Provided by the client:	
Revised MR v1.1 dated 02-November-2024	
Information Verified by Lead Assessor:	
Revised monitoring report version 1.1 dated 02-November-2024	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-November-2024
In Section 1.8 of the revised Monitoring Report (MR) version 1.1, dated 02-November-2024, the format of the geo-coordinates for both project activity instances is now consistent. Additionally, the previously noted error in the longitude has been corrected and aligns with the actual location observed during the site visit. This correction is deemed appropriate and is accepted. CL#2 is CLOSED.	

Date:	17-October-2024		Raised by:	Vikas Bankar	
Type:	CL	Number:	#3	Reference:	Section 1.11.3 of MR, v1,
Lead Assessor Comment:					
Project Proponent has answered "NO" to the question, <i>“Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?”</i> . Could you kindly clarify how this response is appropriate, considering that the project activity directly impacts the Scope 2 emissions of the customers? Since the district cooling system supplies chilled water to customers and reduces energy consumption, it lowers the emissions footprint associated with their electricity usage.					
Client’s Response :				Date : 02-November-2024	
The section 1.11.3 has been corrected in the revised MR. Tabreed is supplier of district cooling, and the project activity reduces the emissions. However, Tabreed has not published in any product manual or on website on specific project instances and claiming any reductions in scope 3 emissions.					
Documentation Provided by the client:					
Revised MR version 1.1 dated 02-November-2024					
Information Verified by Lead Assessor:					
Revised monitoring report version 1.1 dated 02-November-2024					
The reasoning for not Acceptance or Acceptance and Close Out:				Date: 04-November-2024	
				Reopened: 10-March-2025	
The Project Proponent (PP) has now accurately included responses regarding Supply Chain (Scope 3) emissions in Section 1.11.3 of the revised Monitoring Report (MR) version 1.1, dated 02-November-2024. This revision is deemed appropriate and is therefore accepted.					
CL#3 is CLOSED.					
CL#3 is reopened on 10-March-2025					
In Section 1.11.3 of the Monitoring Report (MR), it is stated that the project activities impact the emissions footprint of goods and services within a supply chain. However, there is no evidence of a					

public statement issued by the Project Proponent (PP) or its authorized representative, as required by paragraph 3.24.7 of the VCS Standard v4.7. Please provide clarification.	
Client's Response :	Date : 09-April-2025
The public disclosure is published in the Tabreed website https://www.tabreed.ae/sustainability/ in the downloads as 'Declaration on Supply Chain (Scope 3) Emissions'	
Documentation Provided by the client:	
1. Link of website https://www.tabreed.ae/sustainability/	
Information Verified by Lead Assessor:	
The public disclosure is published on Tabreed website https://www.tabreed.ae/sustainability/ Revised monitoring report version 1.5 dated 10-April-2025.	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 14-April-2025
The project proponent has now published a disclosure stating that " <i>Carbon credits may be issued through Verified Carbon Standard project ID 4519 - TABREED DISTRICT COOLING GROUP PROJECT for the greenhouse gas emission reductions or removals associated with National Central Cooling Company PJSC -Tabreed's cooling/chilled water produced through the district cooling plants.</i> " on their website https://www.tabreed.ae/sustainability/ . Also, PP has done the necessary corrections in section 1.11.3 of the revised monitoring report in line with the requirement of paragraph 3.24.7 of the VCS Standard v4.7. This is found appropriate, and it is accepted.	
CL#3 is CLOSED.	

Date:	17-October-2024	Raised by:	Vikas Bankar
Type:	CL	Number:	#4
Reference:	Section 1.12 of MR, v1,		
Lead Assessor Comment:			
Please clarify why MR completion guidance has not been followed when populating the information in the columns titled "Current Project Contributions" and "Contributions over Project Lifetime" in Table 1, included in Section 1.12 of the report.			
Client's Response :	Date : 02-November-2024		
The section on SDG has been revised to include Current project contributions and contributions over the project lifetime.			
Documentation Provided by the client:			
Revised MR version 1.1 dated 02-November-2024			
Information Verified by Lead Assessor:			
Revised MR version 1.1 dated 02-November-2024			
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-November-2024		

Please provide clarification on the appropriateness of the SDG 13 contributions referenced over the project's lifetime in accordance with MR completion guidelines. Additionally, kindly provide evidence supporting the new job creations cited under the SDG 8.3 contribution. Please also clarify why this evidence is not included in the appendices of the revised MR. CL#4 is OPEN.	
Client's Response :	Date: 07-November-2024
The SDG 13 contributions are updated. SDG 8.3 has been excluded in the absence of the availability of supporting documents.	
Documentation Provided by the client:	
Revised MR version 1.2 dated 07-November-2024	
Information Verified by Lead Assessor:	
Revised MR version 1.2 dated 07-November-2024	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 08-November-2024
PP has submitted the revised monitoring report version 1.2 with SDG 8.3 removed from section 1.12 of MR in the absence of the evidence. This is found acceptable. CL#4 is CLOSED.	

Date:	17-October-2024	Raised by:	Vikas Bankar
Type:	CL	Number:	#5
Reference:	Section 2.4 of MR, v1		
Lead Assessor Comment:			
Inconsistency in Water Source (Sections 1.12 and 2.4) Section 2.4 refers to water consumption from a seawater desalination plant, while Section 1.12 states that the project activity will be using grey water. Could you please clarify this inconsistency observed?			
Client's Response :	Date : 02-November-2024		
The plants are designed for desalinated water and treated sewage effluent (TSE)/grey water. However, TSE is not available presently and therefore desalinated water is used. The section is updated to include only desalinated water.			
Documentation Provided by the client:			
Revised MR version 1.1 dated 02-November-2024			
Information Verified by Lead Assessor:			
Revised MR version 1.1 dated 02-November-2024			
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-November-2024		
PP has now removed the statement about the usage of grey water from section 1.12 of the revised MR version 1.1 dated 02-November-2024 as it is not implemented at this stage. This is found appropriate, and it is accepted. CL#5 is CLOSED.			

Date:	17-October-2024	Raised by:	Vikas Bankar
Type:	CL	Number:	#6
Reference:	Section 4.2 of MR v1		

Lead Assessor Comment:	
Please clarify why is the latest available Transmission and Distribution (T&D) losses not considered, given that TDLB,y, and TDLy are designated as monitored parameters.	
Client's Response :	Date : 02-November-2024
The values used are the latest available (2021). The values will be updated when the new data is available. Furthermore, in the registered PD, the monitoring frequency for this data is 5 years.	
Documentation Provided by the client:	
NA	
Information Verified by Lead Assessor:	
Clarification provided by the Project Proponent	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-November-2024 Reopened: 10-March-2025
The clarification provided by the Project Proponent is deemed appropriate, as alternative sources for transmission and distribution of loss data available publicly are older than 2021. Additionally, in accordance with the registered VCS Project Description, these values are valid for use for up to five years. Therefore, it is accepted. CL#6 is CLOSED. CL#6 is reopened on 10-March-2025 Please clarify why is the latest available Transmission and Distribution (T&D) losses not considered, in line with the CDM TOOL 05.	
Client's Response :	Date : 09-April-2025
The emission reduction calculation is updated with the new values based on Department of energy '2022 Annual Technical Report' Page 31 for transmission losses (2.51%) and page 41 for distribution losses (6%)	
Documentation Provided by the client:	
1. 2022 Annual technical report 2. Updated emission reduction calculations 3. Updated Monitoring report version 1.5	
Information Verified by Lead Assessor:	
• Annual Technical Report 2022 available https://doe.gov.ae/-/media/Project/DOE/Department-Of-Energy/Media-Center-Publications/2022-Annual-Technical-Report-ENG.pdf (last accessed on 14-April-2025) • Revised emission calculation sheet for PAI01 and PAI02 • Revised monitoring report version 1.5 dated 10-April- 2025	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 14-April-2025
PP has updated the transmission and distribution loss in line with Annual Technical Report 2022 available https://doe.gov.ae/-/media/Project/DOE/Department-Of-Energy/Media-Center-Publications/2022-Annual-Technical-Report-ENG.pdf . However, it is observed that Annual Technical Report is also available year 2023. Please clarify why is the T&D loss from the latest technical report 2023 is not applied.	

Client's Response :	Date : 14-April-2025
The emission reduction calculation is updated with the new values based on Department of energy '2023 Annual Technical Report' Page 28 for transmission losses (2.04%) and page 38 for distribution losses (7.2%)	
Documentation Provided by the client:	
1. 2023 Annual technical report 2. Revised emission reduction sheet 3. Revised monitoring report version 1.6 dated 14-April-2025	
Information Verified by Lead Assessor:	
Revised ER calculation sheet for PAI 01 and 02	
Revised Monitoring Report version 1.6 dated 14-April 2025	
Annual Technical Report from DOE 2023	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 17-April-2025
PP has updated the transmission and distribution loss in line with Annual Technical Report 2023 available at https://doe.gov.ae/-/media/Project/DOE/Department-Of-Energy/Media-Center-Publications/technical-reporten.pdf (last accessed on 17-April-2025). This is found to be appropriate, and it is accepted. CL#6 is CLOSED.	

Date:	17-October-2024	Raised by:	Vikas Bankar		
Type:	CAR	Number:	#7	Reference:	Section 5 of MR v1
Lead Assessor Comment:					
1. Emission Calculations Up to December 2023 Section 5 of the report calculates emissions and emission reductions only up to December 2023. Could you please clarify why these calculations do not extend to July 2024, as expected for this reporting period? 2. Leakage Emissions from January to July 2024 In Section 5.4, the table reports leakage emissions as "zero" for the period from January to July 2024. Could you clarify the rationale behind this reporting and confirm if this is accurate? 3. Missing Data for Project Activity Instance 2 The table for the vintage period does not include emission reduction data for Project Activity Instance 2. Please clarify this omission and provide the relevant data.					
Client 's Response :			Date : 02-November-2024		
1. The calculations are extended covering the full monitoring period. The MR is updated. 2. The leakage emissions have 3 main contributors i.e. water use, refrigerant leakage and scrapping of baseline equipment. The water uses and scrapping of baseline emissions are used only one time as per the methodology and during Jan-July 2024, no refrigerant leakage occurred. Therefore, the leakage emissions are zero. 3. All the data for PA02 is included in the revised MR.					
Documentation Provided by the client:					

Revised MR version 1.1 dated 02-November-2024	
Information Verified by Lead Assessor:	
Revised MR version 1.1 dated 02-November-2024	
Reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-November-2024
1. Section 5 of the revised MR now accurately reflects baseline emissions, project emissions, leakage emissions, and emission reduction calculations over the complete monitoring period. This is deemed appropriate and has been accepted. 2. It is confirmed that the maximum designed quantity has been considered for leakage emission calculations, corresponding to the volume of the cooling tower sump as verified from the design document. This approach aligns with paragraph 46 of the applied methodology AM0117, version 2.0. 3. The table for the vintage period now includes the emission reduction data for project activity instance 2 in the revised MR version 1.1, dated 02-November-2024. This update is appropriate and has been accepted. CAR #7 is CLOSED.	

Date:	17-October-2024	Raised by:	Vikas Bankar		
Type:	CL	Number:	#8	Reference:	Section 3.3 of MR v1
Lead Assessor Comment:					
Registered VCS PD mentioned that “The local stakeholder consultation will be conducted before inclusion of any new project instance.” Please confirm if stakeholder consultation was carried out before the inclusion of project activity instance 2.					
Client’s Response :				Date : 02-November-2024	
The stakeholder consultation happened on 19-September-2023 during validation which was before the inclusion of the new project instance. Furthermore, the project participant has a continuous grievance procedure. The project proponent will conduct another stakeholder consultation before including the new project instance.					
Documentation Provided by the client:					
1. Stakeholder consultation presentation and other documents					
Information Verified by Lead Assessor:					
Clarification provided by the project proponent					
The reasoning for not Acceptance or Acceptance and Close Out:				Date: 04-November-2024 Reopened: 10-March-2025	
The Project Proponent clarified that the stakeholder consultation was conducted during the validation of project activity instance 1, with supporting evidence verified. Given that both plants are of similar capacity and are geographically proximate, the stakeholders for both project activity instances are considered to be the same. As a result, PP did not conduct a separate stakeholder consultation for project activity instance 2. This deviation for project activity instance 2 is accepted. Also, during the onsite visit, the project proponent was unable to provide a contractual document verifying the requirement of VCU ownership as mentioned in section 3.3 of the MR. However, it was observed that the project proponent has solely invested in and developed the project activity, with the end user having no role or involvement in the project activity itself. The project proponent further clarified that they include a clause in the customer service agreement specifying the proponent's VCS ownership. Based on this clarification, the VCS ownership of the project proponent is deemed acceptable. However, FAR#11 is raised for the verification team in future monitoring cycles. There, CL#8 is CLOSED. CL#8 is reopened on 10-March-2025 The eligibility criteria for the inclusion of Project Activity Instance 2 (PAI2) have not been met as per the registered VCS Project Description (PD). Please provide clarification. Additionally, the date of the stakeholder consultation conducted for Project Activity Instance 1 (PAI1) is inconsistent with the registered VCS PD and the validation report. Kindly clarify.					
Client’s Response :				Date : 09-April-2025	

The stakeholder consultation has not been conducted for the PAI2 as the PAI2 started before the project registration. Project's focal point has applied for the exemption for this requirement from the verra secretariat and received the exemption approval to conduct the stakeholder consultation before second verification. Therefore, the eligibility conditions meet the requirements of registered VCS PD. The PAI 1 stakeholder date is corrected in the MR.	
Documentation Provided by the client:	
1. Approved exemption from verra dated 08th April 2025 2. Revised Monitoring report version 1.5	
Information Verified by Lead Assessor:	
• Approved exemption from VERRA dated 08-April-2025 • Revised monitoring report version 1.5 dated 10-April-2025	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 14-April-2025
PP has now submitted approved exemption from VERRA dated 08-April-2025 with respect eligibility criteria 8 as per the registered VCS PD. VERRA allowed the project proponent to carry out stakeholder consultation prior to second verification; hence it is accepted. This requirement is now added to FAR#11 so that the verification team shall ensure that the stakeholder consultation for PAI 02 is completed before the start of second verification. Also, PP has now corrected the date of date of the stakeholder consultation conducted for PAI 01 as the 19-September-2023 which is consistent with the VCS PD, validation report and stakeholder consultation documents available at the time project registration. This is found appropriate, and it is accepted. CL#8 is CLOSED.	

Date:	17-October-2024	Raised by:	Vikas Bankar
Type:	CAR	Number:	#9
Reference:	Section 3.3 of MR v1, ER calculation sheets		
Lead Assessor Comment:			

1. Section 4.2 of the monitoring report does not include accurate meter numbers, specifications, and calibration details for the BTU and energy meters. Could you please clarify why these details have not been provided for both plants? 2. Additionally, Section 4.2 mentions that the cooling energy generated at both ends, i.e., DCP and the customer end, will be compared, and the lower value of TRh generation will be used in the calculation to ensure conservativeness. However, this approach is not demonstrated in the emission reduction calculation sheet. 3. Similarly, the report indicates that electricity consumption measured at PMU meters and ADDC invoices will be compared, with the higher value used for calculations. This approach is also not reflected in the emission reduction calculation sheet. 4. A minor deviation is observed in the reported values of cooling output and electricity consumption for both the project activity instances when compared with SAP extracts. Please clarify the deviations observed.	
Client's Response :	Date: 02-November-2024
1. The MR is revised to include the details of all the meters. All the meters are revenue-grade meters and calibrated as per national standards. All the calibration records have been submitted. 2. The emission reduction sheets have been updated to include the cooling generated and cooling used by the consumers and lower value has been used in the emission reduction calculation conservatively. 3. Electricity consumption is compared with the ADDC invoices and higher value has been used conservatively. The emission reduction sheet has been revised. 4. The deviation in the values has been corrected in the revised MR.	
Documentation Provided by the client:	
Revised MR version 1.1 dated 02-November-2024 Calibration records of meters Revised emission reduction calculation sheets for PA01 and PA02 ADDC invoices for PA01 and PA02	
Information Verified by Lead Assessor:	
Revised MR version 1.1 dated 02-November-2024 Calibration records of meters Revised emission reduction calculation sheets for PA01 and PA02 in reference to revised MR version 1.1 dated 02-November-2024 ADDC invoices for PA01 and PA02	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-November-2024

1. Section 4.2 of the revised MR now includes meter details, such as serial numbers. However, the specifications—such as make, model, and accuracy class—are not mentioned. Please clarify. Additionally, the BTU meter and internal electricity consumption meters are under the control of the project proponent. Please explain why calibration details are not included in the monitoring report. **Open.**
2. The revised ER sheets, as per MR version 1.1 dated 02-November-2024, now demonstrate a comparison between the cooling output measured at DCP and the customer, with the lower of the two values used for baseline emission calculations. This approach is appropriate and thus accepted.
3. Similarly, the higher electricity consumption values measured at Tabreed's internal meters and ADDC invoices are used for project emission calculations. This approach is found to be appropriate, conservative, and consistent with the registered monitoring plan, and is therefore accepted.
4. The revised ER sheet and MR are now consistent with monitored SAP data and ADDC invoices. However, the following clarification is required:
 - a) The BTU meter was replaced at the plant on 01-June-2024; however, the cooling output for June 1 has not been considered in the calculations. Please refer to the extract from the ER sheet for AD21 below. **Open.**

Cooling production in the plant (TRh)				
Date	Meter reading (TRH)	Total consumption (TRH)		
3-May-24	33,983,253.030			
1-Jun-24	37,146,393.630	3,163,141	Meter change	
2-Jun-24	116,490.515			
31-Jul-24	9,062,051.053	8,945,561		
		12,108,701		

- b) During the site visit, three ADDC incomers were observed. The ADDC invoices provided align with the meters (Serial Nos. E2T12D000153 and E2T12D000155) for incomers 1 and 3. However, the meter installed at incomer 2 (Serial No. E2T12D000154) does not appear in the ADDC invoice. Instead, a new meter number (Serial No. I1C14D005987) is reflected in the invoice, and its consumption has been included in the project emission calculations. Please clarify this observed inconsistency. **Open.**

Client's Response:
Date: 07-November-2024

- 1, Meter numbers and make are included in the revised monitoring report.
2. The meter change report is submitted, and the change readings have been included in the calculation sheet.
3. The meter of incomer 2 (Serial No. E2T12D000154) has no load and therefore no ADDC bill has been received for this meter. The meter number (Serial No. I1C14D005987) is for booster pump station consumption. This meter on SAP is under the AD-021 plant and therefore consumption is included in the project emission calculations.

Documentation Provided by the client:

1. Meter change report for AD021 2. Revised calculation sheet 3. Revised Monitoring Report version 1.2 dated 07-November-2024	
Information Verified by Lead Assessor:	
Revised ER calculation sheet Revised MR version 1.2 dated 07-November-2024 Acknowledgment report for meter change in AD21 dated 01/06/2024	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 08-November-2024
1. It is observed that the PP has provided the meter details along with the calibration details in section 4.2 of the revised MR version 1.2 dated 07-November-2024. However. a) Calibration certificates are not provided except the factory calibration certificates for BTU meters installed in DCPs. Please clarify. b) The calibration details provided don't cover the entire monitoring period. Please clarify. 4. a) PP has provided the revised ER calculation sheet in reference to revised MR version 1.2 where the initial reading of the new meter is appropriately considered in the cooling output calculations; hence it is accepted. 4. b) The clarification provided by PP was found acceptable and consistent with the site visit observations. During the site visit, it was observed that incomer 2 is a standby feeder and wasn't in use during the current monitoring period; hence accepted. Also, considering the consumption of booster pump stations is the conservative approach.	
Client's Response:	Date : 18-November-2024
Calibration certificates have been provided for BTU meters. However, the calibration certificates are after the monitoring period (meters are working within the acceptable range) and therefore the adjustment is applied by reducing the values by 0.2% (accuracy of BTU meters). This is as per the CDM EB guidance.	
Documentation Provided by the client:	
1. Revised emission reduction sheet 2. Revised Monitoring report v 1.3	
Information Verified by Lead Assessor:	
Revised emission reduction sheet Revised Monitoring report v 1.3 Calibration certificates of BTU meters at SeaWorld DCP and ETS room dated 04-June-2024	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 18-November-2024

The project proponent has provided calibration certificates for the BTU meters at the SeaWorld DCP and ETS room, dated 04-June-2024, with satisfactory calibration results. However, these certificates do not cover the monitoring period from 25-May-2023 to 03-June-2024, as the previous calibration was valid only until 20-April-2023. To address this gap, the project proponent applied the maximum permissible error to the entire monitoring period in line with para 4a) of Guidelines for assessing compliance with the calibration frequency, version 01/40/. The assessment team considers this a suitably conservative approach and has, therefore, accepted it.

CAR#9 is CLOSED.

Date:	03-December-2024		Raised by:	Vikas Bankar	
Type:	CL	Number:	#10	Reference:	Final completeness check comments
Lead Assessor Comment:					
In MR: please clarify on below points - Page numbers are not reflecting in MR - On the last page of the MR the page titled "APPENDIX X: <TITLE OF APPENDIX>" should be removed, since as per the template instructions: <i>"Delete this appendix (title and instructions) where no appendix is required."</i> - In section 2.4.1, in the table, empty row to be deleted - In MR under section 4.3 there is an error in the name of the methodology. - The format of the dates in the Monitoring Report (MR) are not consistent with the template filling instructions. In ER sheet: - Monitoring Period start date is 25-May-2023, However in ER sheet <i>"Emission Reduction calculation_First Verification_instance 1_sp2"</i>, under tab – "Sea World Data", the start date of the monitoring period is mentioned as 26/05/2023. Please clarify.					
Client's Response :				Date : 04-December-2024	
- Page numbers are included in the MR - The "APPENDIX X: <TITLE OF APPENDIX>" has been removed from MR. - In the section 2.4.1, the empty row is deleted - The Methodology name is corrected in the section 4.3 - The format of the dates is updated in the monitoring report (MR) In ER sheet: Monitoring start date is 25-May-2023 which is the date of commissioning of the Sea World project (project instance 1). The data for 25-May-2023 was unavailable and therefore the emission reduction calculation started from the 26-May-2023 without changing the crediting period (i.e. no emission reductions have been claimed for 25-May-2023). The emission reduction calculation sheet has been updated and date is revised.					
Documentation Provided by the client:					
1. Revised MR version 1.4 date 04-December-2024					

2. Revised Emission Reduction Calculation sheet	
Information Verified by Lead Assessor:	
Revised MR version 1.4 dated 04-December-2024 Revised Emission Reduction Calculation sheet relevant to revised MR dated 04-December-2024	
The reasoning for not Acceptance or Acceptance and Close Out:	Date: 04-Decmeber-2024
Monitoring Report (MR): 1. The revised MR dated 04-December-2024 now includes page numbers. This point is considered closed. 2. The blank appendix on the last page of the revised MR dated 04-December-2024 has been removed and is found appropriate. 3. The empty row in the table in Section 2.4.1 has been deleted and is found appropriate. 4. The incorrect methodology name, earlier mentioned as ACM0017, has been corrected to AM0017. This correction is found appropriate and accepted. 5. Dates throughout the monitoring period have been corrected to the DD-Month-YYYY format in the revised MR dated 04-December-2024. This is also accepted. Emission Reduction (ER) Sheet: 6. The Project Participant (PP) clarified that while the monitoring period started on 25-May-2023, data recording commenced on 26-May-2023. This clarification has been incorporated into the revised ER calculation sheet for PA 01. CL#10 is now CLOSED.	

Date:	10-December-2024 Revised on 14-April-2025	Raised by:	Vikas Bankar		
Type:	FAR	Number:	#11	Reference:	PA inclusion eligibility conditions
Lead Assessor Comment:					
Referring to CL#8 above, FAR#11 will remain open for the rest of the crediting period. The verification team, in future monitoring periods, shall verify that - Local stakeholder consultation is completed for all future project activity instances added to the grouped project. - Local stakeholder consultation for PAI 02 is completed before the start of second verification of the project activity. - The clause clarifying the VCU ownership has been added in the new customer service agreement for the existing and future project activity instances included in in the grouped project activity Therefore, FAR#11 is open.					
Client's Response :				Date :	
Documentation Provided by the client:					
Information Verified by Lead Assessor:					

The reasoning for not Acceptance or Acceptance and Close Out:	Date:
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APPENDIX 4: ABBREVIATIONS

Abbreviation	Full Form
ADDC	Abu Dhabi Distribution Company
ADNOC	Abu Dhabi National Oil Company
ADDED	Abu Dhabi Department of Economic Development
ADIS	Abu Dhabi Industrial Strategy
ADSSC	Abu Dhabi Sewerage Services Company
ADPC	Abu Dhabi Ports Company
ADQ	Abu Dhabi Quality and Conformity Council
BTU	British Thermal Unit
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
DCT	Department of Culture and Tourism
DCP	District Cooling Plant
DNV	DNV Business Assurance India Private Limited
DoE	Department of Energy
EAD	Environment Agency Abu Dhabi
ER	Emission Reduction
EWEC	Emirates Water and Electricity Company
FAR	Forward Action Request
GHG	Greenhouse Gas
IFI	International Financial Institutions
IRENA	International Renewable Energy Agency
ISO	International Organization for Standardization
JCI	Johnson Controls Inc.
MP	Monitoring Plan
MR	Monitoring Report
MS	Microsoft
NOC	No Objection Certificate
O&M	Operations and Maintenance
PAI	Project Activity Instance
PCS	Project Control System
PD	Project Description

Abbreviation	Full Form
PP	Project Proponent
QA/QC	Quality Assurance/Quality Control
SAP	Systems, Applications, and Products in Data Processing
SDG	Sustainable Development Goal
SEER	Seasonal Energy Efficiency Ratio
T&D	Transmission and Distribution
T&C	Testing and Commissioning
TDL	Transmission and Distribution Loss
TSE	Treated Sewage Effluent
UAE	United Arab Emirates
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit
VVB	Validation and Verification Body

APPENDIX 5: COMPETENCY CERTIFICATES



Individual Competency statement

NAME: VIKAS VILASRAO BANKAR

has been qualified as per ICP_1_GHG_CMP_Competency Management Procedure which is in accordance with ISO 14066:2011

Qualified for GHG Programme

Voluntary Carbon Standard (VCS)	☒	Gold Standard (GS)	☐
Global Carbon Council (GCC)	☒	Other, specify _____	☐

For Management Functions		For Assessment Team Roles	
Quality Manager	☐	Team Leader	☒
Technical Manager	☒	Validator / Verifier	☒
GHG Manager	☐	Validator / Verifier (Trainee)	☐
HR Manager	☐	Local Expert, (Country: IN, UAE)	☒
Internal Auditor	☐	Technical Reviewer	☒
		Financial Expert	☐

SS	For Technical Expert Roles		SS	For Technical Expert Roles	
1	TA 1.1 – Thermal Energy Generation	☒	9	TA 9.1 – Aluminium and magnesium production	☐
	TA 1.2 – Renewables	☒		TA 9.2 – Iron, steel, and Ferro-alloy production	☐
2	TA 2.1 – Energy Distribution	☒	10	TA 10.1 – Fugitive emissions from oil and gas	☐
3	TA 3.1 – Energy Demand	☒	13	TA 13.1 – Solid waste and wastewater	☐
			13	TA 13.2 – Manure	☐
4	TA 4.1 – Cement and Lime Production	☐	14	TA 14.1 – Afforestation and reforestation	☐
6	TA 6.1 – Construction	☐	15	TA 15.1 - Agriculture	☐
7	TA 7.1 - Transport	☐	16	TA 16.1 – Carbon capture and storage	☐

Prepared by: Anjana Sharma	Approved by: Anjana Sharma
Date: 17/06/2023	Date: 17/06/2023 
Valid up to: 16/06/2025	



Individual Competency statement

NAME: HIMANSHU BABBAR

has been qualified as per ICP_1_GHG_CMP_Competency Management Procedure which is in accordance with ISO 14066:2011

Qualified for GHG Programme

Voluntary Carbon Standard (VCS)	☐	Gold Standard (GS)	☐
Global Carbon Council (GCC)	☐	Other, specify _____	☐

For Management Functions

Quality Manager	☐	Team Leader	☐
Technical Manager	☐	Validator / Verifier	☐
GHG Manager	☐	Validator / Verifier (Trainee)	☒
HR Manager	☐	Local Expert, (Country: India)	☒
Internal Auditor	☐	Technical Reviewer	☐
		Financial Expert	☐

For Assessment Team Roles

SS	For Technical Expert Roles		SS	For Technical Expert Roles	
1	TA 1.1 – Thermal Energy Generation TA 1.2 – Renewables	☐ ☐	9	TA 9.1 – Aluminium and magnesium production TA 9.2 – Iron, steel, and Ferro-alloy production	☐ ☐
2	TA 2.1 – Energy Distribution	☐	10	TA 10.1 – Fugitive emissions from oil and gas	☐
3	TA 3.1 – Energy Demand	☐	13	TA 13.1 – Solid waste and wastewater TA 13.2 – Manure	☒ ☐
4	TA 4.1 – Cement and Lime Production	☐	14	TA 14.1 – Afforestation and reforestation	☐
6	TA 6.1 – Construction	☐	15	TA 15.1 - Agriculture	☐
7	TA 7.1 - Transport	☐	16	TA 16.1 – Carbon capture and storage	☐

Prepared by: Vikas Bankar

Date: 18/12/2023

Valid up to: 18/12/2025



Approved by: Anjana Sharma

Date: 19/12/2023





Individual Competency statement

NAME: Shaikha Alhammadi

has been qualified as per ICP_1_GHG_CMP_Competency Management Procedure which is in accordance with ISO 14066:2011

Qualified for GHG Programme

Voluntary Carbon Standard (VCS)	☐	Gold Standard (GS)	☐
Global Carbon Council (GCC)	☐	Other, specify _____	☐

For Management Functions

For Assessment Team Roles

Quality Manager	☐	Team Leader	☐
Technical Manager	☐	Validator / Verifier	☐
GHG Manager	☐	Validator / Verifier (Trainee)	☐
HR Manager	☐	Local Expert, (Country: IN, UAE)	☐
Internal Auditor	☐	Local Language Expert, (Country & Language: UAE/ Arabic)	☒
		Technical Reviewer	☐
		Financial Expert	☐

SS	For Technical Expert Roles	SS	For Technical Expert Roles
1	TA 1.1 – Thermal Energy Generation TA 1.2 – Renewables	9	TA 9.1 – Aluminium and magnesium production TA 9.2 – Iron, steel, and Ferro-alloy production
2	TA 2.1 – Energy Distribution	10	TA 10.1 – Fugitive emissions from oil and gas
3	TA 3.1 – Energy Demand	13	TA 13.1 – Solid waste and wastewater TA 13.2 – Manure
4	TA 4.1 – Cement and Lime Production	14	TA 14.1 – Afforestation and reforestation
5	TA 5.1 – Chemical Industry TA 5.2 - Caprolactam, nitric and adipic acid	15	TA 15.1 - Agriculture
6	TA 6.1 – Construction	16	TA 16.1 – Carbon capture and storage
7	TA 7.1 - Transport		

Prepared by: Shilpa Swarnim

Approved by: Anjana Sharma

Date: 08 October 2024

Date: 08 Oct 2024

Valid up to: 07 October 2026



Individual Competency statement

NAME: VENKATA RAMAN KAKARAPARTHI

has been qualified as per ICP_1_GHG_CMP_Competency Management Procedure which is in accordance with ISO 14066:2011

Qualified for GHG Programme

Voluntary Carbon Standard (VCS)	☒	Gold Standard (GS)	☐
Global Carbon Council (GCC)	☐	Other, specify _____	☐

For Management Functions

Quality Manager	☐
Technical Manager	☐
GHG Manager	☐
HR Manager	☐
Internal Auditor	☐

For Assessment Team Roles

Team Leader	☒
Validator / Verifier	☐
Validator / Verifier (Trainee)	☐
Local Expert, (Country: IN)	☒
Technical Reviewer	☒
Financial Expert	☐

SS	For Technical Expert Roles	SS	For Technical Expert Roles
1	TA 1.1 – Thermal Energy Generation TA 1.2 – Renewables	9	TA 9.1 – Aluminium and magnesium production TA 9.2 – Iron, steel, and Ferro-alloy production
2	TA 2.1 – Energy Distribution	10	TA 10.1 – Fugitive emissions from oil and gas
3	TA 3.1 – Energy Demand	13	TA 13.1 – Solid waste and wastewater TA 13.2 – Manure
4	TA 4.1 – Cement and Lime Production	14	TA 14.1 – Afforestation and reforestation
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